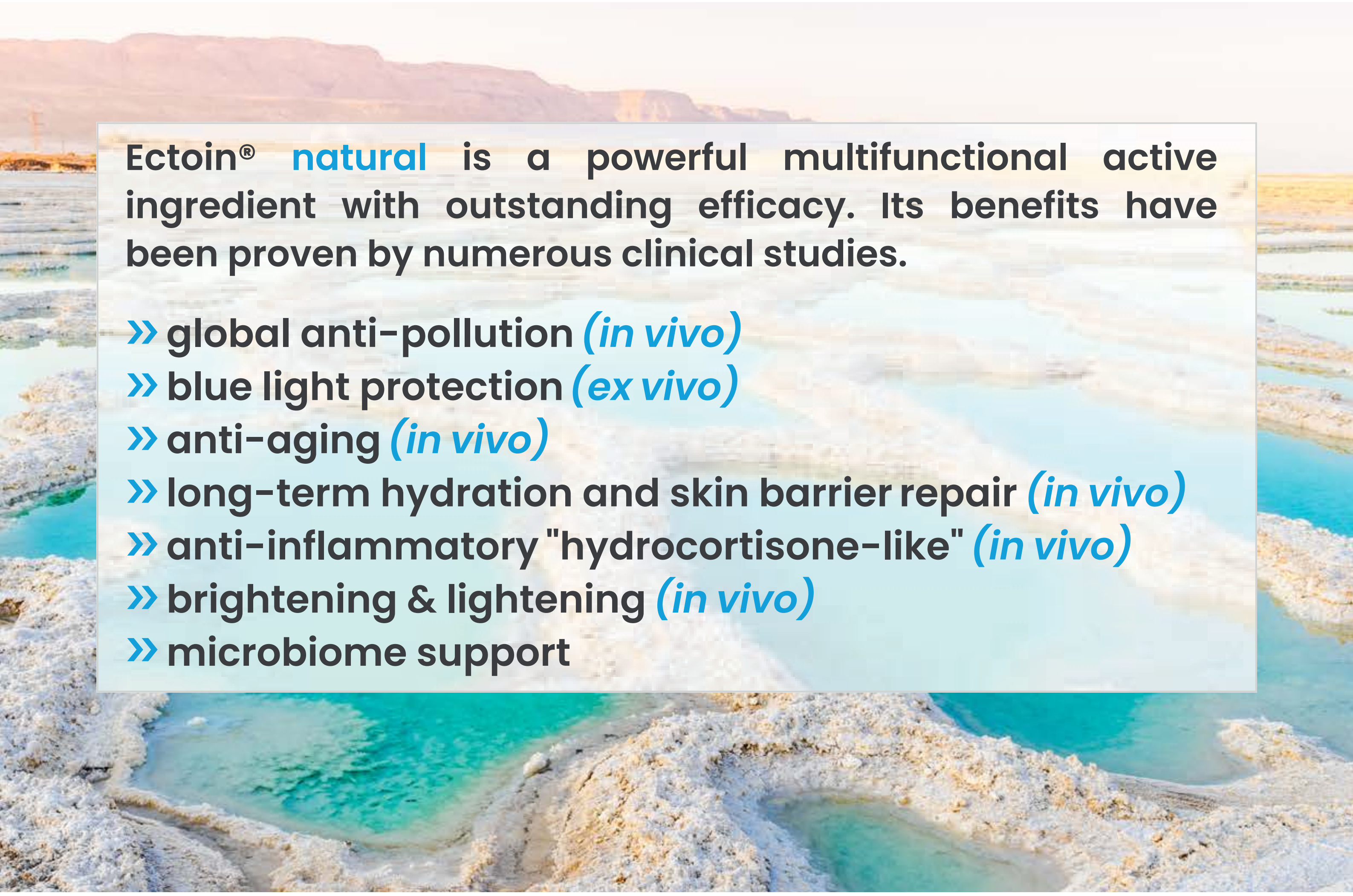
Ectoin® natural

ULTIMATE PROTECTION & REPAIR

100% PURE & NATURAL.
THE ORIGINAL: only made by bitop

Ectoin® natural is a powerful multifunctional active ingredient with outstanding efficacy. Its benefits have been proven by numerous clinical studies.

- » global anti-pollution *(in vivo)*
- » blue light protection *(ex vivo)*
- » anti-aging *(in vivo)*
- » long-term hydration and skin barrier repair *(in vivo)*
- » anti-inflammatory "hydrocortisone-like" *(in vivo)*
- » brightening & lightening *(in vivo)*
- » microbiome support



"The Dead Sea": habitat for Ectoin® natural producing extremophilic microorganisms.

MADE IN GERMANY

CHINA APPLICABLE

Approved by ECOCERT RAW MATERIAL

COSMOS APPROVED

Vegan

MADE IN GERMANY
PROTECTED QUALITY

Nature provides the idea
– we add the science.

bitop
Extremolytes for life

TABLE OF CONTENTS

Ectoin® natural
ULTIMATE PROTECTION & REPAIR

Ectoin® natural is a very safe, 100% natural, and multifunctional cosmetic active ingredient with outstanding cell protection and anti-aging properties. Various *in vivo* and *in vitro* studies as well as clinical trials underline the extraordinary efficacy of **Ectoin®** natural.

INCI: Ectoin
Appearance: white powder (water-soluble, clear)

Ectoin® natural was discovered in 1985 as the self-defense and survival substance of microorganisms living in a salt lake in Wadi El Natrun (Egyptian desert).

Ectoin® natural: CELL PROTECTION & REPAIR DESIGNED BY NATURE

Ectoin® **natural** is an amino acid derivate and belongs to the group of Extremolytes. Extremolytes are small stress-protection molecules, which protect extremophilic microorganisms and plants from the lethal and extreme conditions of their habitats like salt lakes, hot springs, arctic ice, the deep sea, or deserts.

The Dead Sea: habitat for extremophilic microorganisms.

Ectoin® natural shows global and powerful protection efficacy. It shields the skin from extrinsic and intrinsic factors of skin aging and cell damage like pollution particles, sunlight (including IR-A and visible light), temperature, and chemical stress.

At the same time, **Ectoin®** natural enhances and restores cell functions, which are responsible for the skin's health and beauty. This global protection and repairing activity leads to long-lasting and visible preservation and restoration of youthful and healthy skin.

MODE OF ACTION

Ectoin® natural
ULTIMATE PROTECTION & REPAIR

Salt lakes and deserts are the most hostile environments on earth. It is astonishing that despite these lethal conditions there has been life for millions of years. Highly adapted bacteria, so-called extremophilic microorganisms, defy dryness, high salinity, and extreme changes in temperature and grow well due to a substance called **Ectoin®** natural.
It protects these small living beings from harmful environmental influences of their habitats. The efficacy of **Ectoin®** natural has been investigated in detail by numerous independent scientific research groups and universities in Germany and worldwide. Today, the fascinating natural protection properties of **Ectoin®** natural are applied in health care, life science, and cosmetics.

SMALL MOLECULE – BIG IMPACT

Ectoin® natural is a highly kosmotropic substance. The small amino acid derivate binds water molecules from its surroundings and creates the so-called **"Ectoin® Hydro Complex"** (figure 1). These complexes then surround cells, enzymes, proteins, and other biomolecules by forming protective, nourishing, and stabilizing hydration shells around them (figure 2).

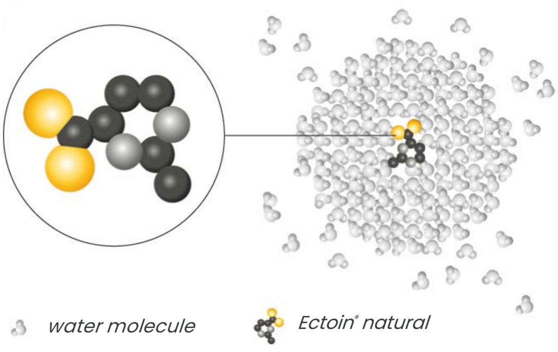


Figure 1: "Ectoin® Hydro Complex".

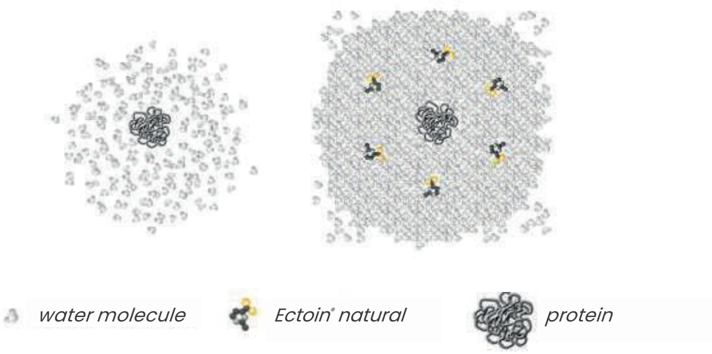


Figure 2: Protection shell around protein.

The **Ectoin®** natural protection mechanism is simple yet effective: The "**Ectoin® Hydro Complexes**" protect and stabilize cell membranes, thus reducing oxidative stress for the cells and the expression of damaging inflammatory genes (stop of cell inflammation).

In addition to that, the fluidity of the cell membranes is enhanced by the "**Ectoin® Hydro Complexes**" resulting in a higher activity of cells, and a boost of the skin's self-defense and repair mechanism. Furthermore, the "**Ectoin® Hydro Complexes**" increase the water content of the skin (figure 3).

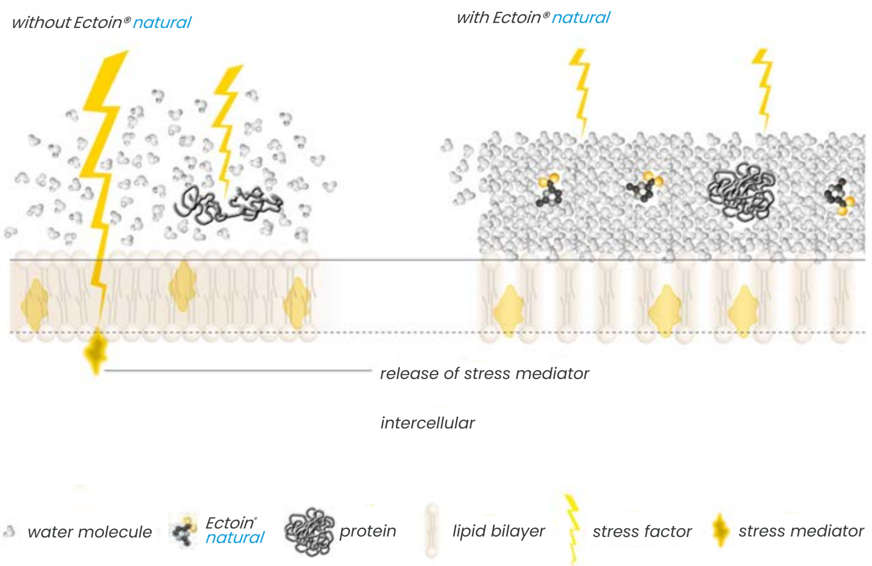


Figure 3: Surface of cell membrane (extracellular space) with and without Ectoin® natural.

Various studies have clearly displayed the most powerful stabilizing properties of **Ectoin®** natural on the integrity of proteins, nucleic acids, biomembranes, and whole cells compared to other solutes and substances.

Ectoin® natural : PROTECTION, PREVENTION, REPAIR AND REGENERATION

The outstanding stabilizing and protective characteristics of **Ectoin®** natural lead to visible and long-term anti-aging results for our skin. Clinical studies have shown a long-lasting enhancement of the skin condition like an increase of elasticity, reduction of wrinkle volume, or skin roughness. By repairing the skin barrier and restoring and regulating the moisture content of the skin, **Ectoin®** natural reduces the TEWL (transepidermal water loss), raises the hydration level, and preserves the skin moisture for seven days without any re-application!

Furthermore, **Ectoin®** natural soothes irritated and damaged skin. Because of its excellent anti-inflammatory properties, the molecule is also used in health care products (OTC medical devices) for the treatment of atopic dermatitis (neurodermatitis) or allergic skin conditions.[3]

ANTI-WRINKLE EFFICACY IN THE CROW'S FEET AREA *(IN VIVO)*

An *in vivo* study was carried out on 10 female volunteers over a period of four weeks. The aim of the study was to analyze the efficacy of 0.5% **Ectoin®** natural with regard to the reduction of wrinkles in the crow's feet area.

Study protocol:

- placebo-controlled *in vivo* study with 10 female volunteers (45-65 years old)
- treatment with cream containing 0.5% **Ectoin®** natural and placebo 2x/day for 4 weeks
- application area: outer corner of the eye (crow's feet area)
- optical measurements of the skin surface by Primos® method on day 0 and day 28
- measured parameters: mean wrinkle depth, deepest wrinkle, wrinkle volume, waveness depth, mean roughness

Table 1: Results of wrinkle reduction (crow's feet area) after four weeks of twice-daily application with 0.5% **Ectoin®** natural cream vs. placebo.

Skin parameter	Cream with 0.5% Ectoin® natural changes in %	Placebo cream changes in %	Cream with 0.5% Ectoin® natural, changes compared to placebo in %
Mean wrinkle depth (µm)	- 19%	+ 13%	- 32%
Deepest wrinkle (µm)	- 3%	+ 14%	- 17%
Volume of the wrinkles (mm^3)	- 21%	+ 19%	- 40%
Waveness depth (Wt) (µm)	- 22%	+ 8%	- 30%
Mean roughness (Rz) (µm)	- 8%	+ 27%	- 35%

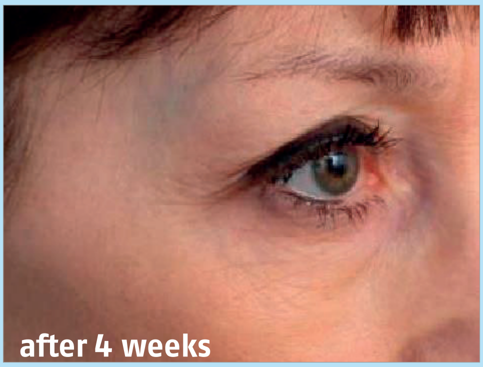
Results

Four-week treatment with 0.5% **Ectoin®** natural led to a remarkable improvement of all wrinkle parameters in the eye area. 100% of the test subjects demonstrated a significant anti-wrinkle effect with **Ectoin®** natural.

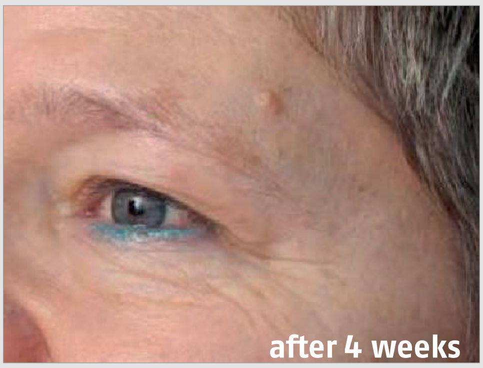
On the contrary, the placebo cream showed no positive efficacy at all. Here, all the wrinkle parameters even led to a deterioration of the respective values.

Anti-wrinkle efficacy in crow's feet area after 4 weeks of 2x/daily application with 0.5% Ectoin® natural cream vs. placebo:

0.5% Ectoin® natural



placebo



ANTI-AGING EFFICACY (CLINICAL STUDY)

Study protocol:

- double-blind, intra-individual, placebo-controlled clinical study
- 24 female volunteers (30-60 years old) with normal, dry, and sensitive skin using cream containing 2% **Ectoin®** natural and placebo cream 2x/day for 28 days
- measured parameters:
    - skin surface parameters (smoothness, wrinkle depth, scaling, roughness) measured by SELS method (Visioscan®) on day 0 and day 28
    - visco-elasticity & biologic elasticity measurement with Cutometer® (in AU) at day 0 and day 28

Results

After four weeks of treatment with **Ectoin®** natural, a visible and significant improvement of the skin microrelief was observed. Deep structures were smoothed. The skin looked healthier and rejuvenated.

100% of the subjects demonstrated a significant effect for all measured parameters.[1] The wrinkle volume was improved by up to 23% after four weeks of treatment with **Ectoin®** natural. Skin scaling was reduced by -76%, roughness by up to -86%. The skin's elasticity was increased.[1]

Skin smoothing and reduction of wrinkle depth after 28 days

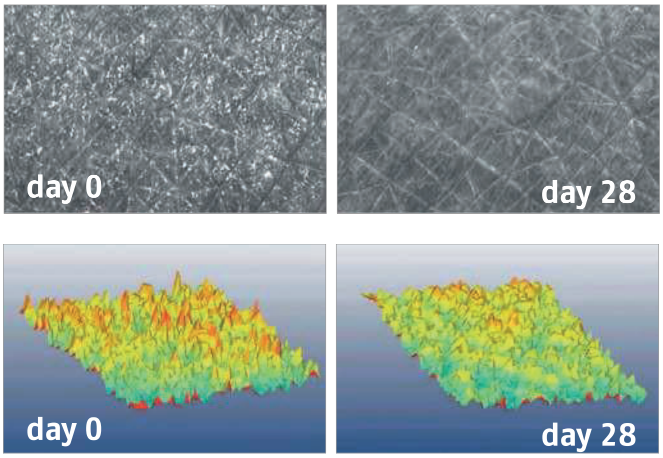
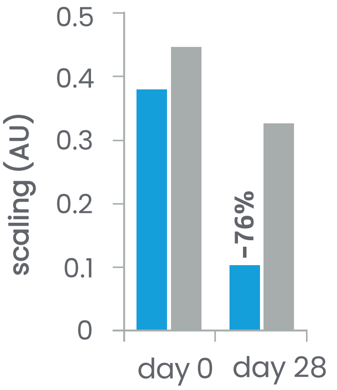


Figure 4: Skin surface measurements by SELS method (Visioscan). Source: Heinrich *et al.*, 2007.

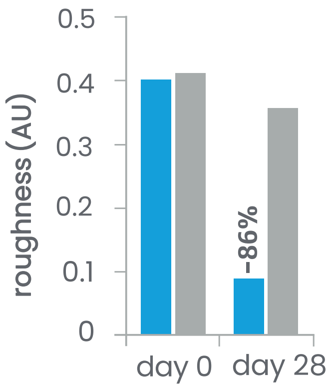
Reduction of skin scaling and roughness after 28 days



■ 2% Ectoin® natural
■ placebo

- 76% scaling

Figure 5: Scaling measured by SELS method (Visioscan).



- 86% roughness

Figure 6: Roughness measured by SLES method (Visionscan).

Increase of skin elasticity

Table 2: Skin elasticity - before and after 28 day treatment with placebo and **Ectoin®** natural vs. untreated control.

Treatment with	Visco-elasticity compared to untreated control
Ectoin® natural	+ 82.4%
Placebo	+ 38.5%

100% PROTECTION OF LANGERHANS CELLS *(IN VIVO)*

Ectoin® natural is proven to 100% protect epidermal immune cells called "Langerhans cells" from damage. In addition to the skin's antioxidant self-defense, these cells help to protect the skin by recognizing antigens and inducing antibody defense responses. Langerhans cells are sensitive to external stress factors. A decreasing number of Langerhans cells is a sign of skin damage and aging.

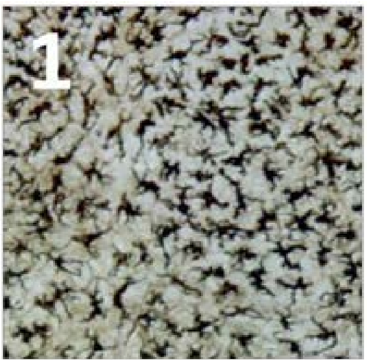
An *in vivo* study impressively demonstrated how **Ectoin®** natural prevents aging and skin damage by protecting the Langerhans cells from UV-induced damage and reduction.[2]

Studyprotocol:

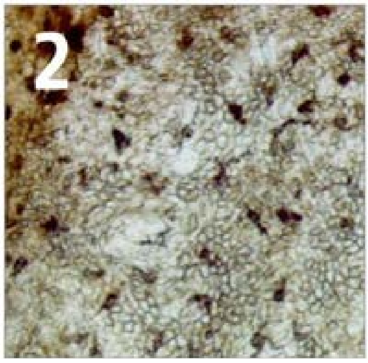
- *in vivo* study, placebo-controlled
- Treatment 2x/day with placebo, 0.3% or 0.5% **Ectoin®** natural for 14 days
- UV-stress (1.5 MED) of human forearm
- skin preparation and histological proof of Langerhans cells

Protection from UV-induced skin damage

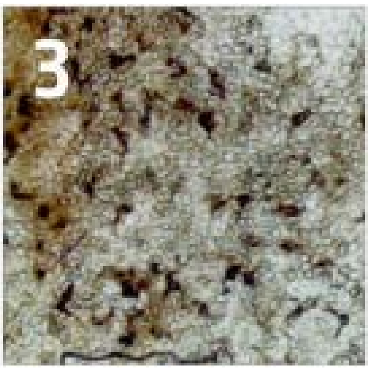
Compare picture 1 with 4.



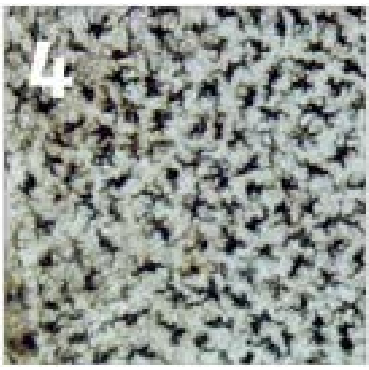
untreated / un-irradiated control



untreated / UV-irradiated



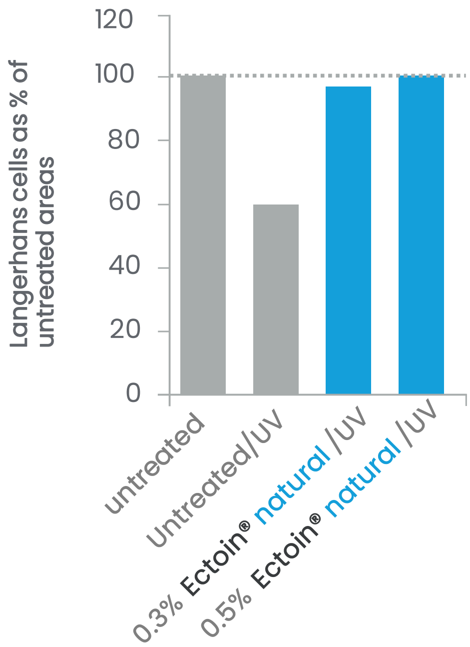
placebo-treated / UV-irradiated



0.5% Ectoin® natural treated / UV-irradiated

Figure 7: Human epidermis under the microscope.
Langerhans cells have been visualized using ATPase staining.

Photographs were taken 48 h after irradiation of skin with 1.5 MED.
Source: Bünger *et al.*, 2001.



100% protection of cells *in vivo*

Figure 8: The amount of vital Langerhans cells in % after UV irradiation on untreated and pre-treated skin with 0.3% and 0.5% Ectoin® natural.

Results

Ectoin® natural protects the amount of Langerhans cells in UV-irradiated human skin and thus reduces UV-induced cellular immune-suppression.[2]

Ectoin® natural prevents photoaging by protecting the Langerhans cells from UV-induced damage.

0.5% **Ectoin®** natural was sufficient to protect 100% of the Langerhans cells. UV-stress did not damage the Langerhans cells in the **Ectoin®** natural treated areas.

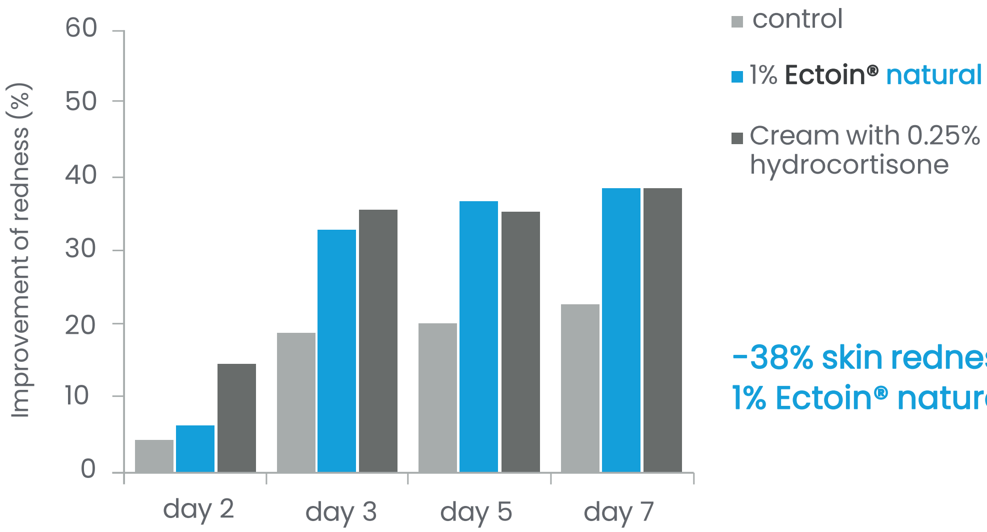
REDUCTION OF SKIN IRRITATION "HYDROCORTISONE-LIKE" (CLINICAL STUDY)

Ectoin® natural is proven to be a side-effect-free replacement for corticosteroids. It is used in health care products (OTC medical devices) for the treatment of eczema, neurodermatitis, allergic and atopic skin. **Ectoin®** natural is safe and approved for the treatment of inflamed and atopic baby skin. Please ask for our medical studies for further information about the use of **Ectoin®** natural in medical devices.

Studyprotocol:

- clinical, placebo-controlled, intra-individual, randomized study
- 20 test persons
- on day 1 skin was stressed for 24 h with 2% SDS solution (sodium dodecyl sulfate) to damage the skin barrier of test persons
- after SDS stress, application of 1% **Ectoin®** natural, 0.25% hydrocortisone ("Ebenol Cream") as well as placebo on stressed skin
- application: 2x/day for 7 days on inner forearms
- measured parameters: TEWL and skin redness

Reduction of skin redness



-38% skin redness with 1% Ectoin® natural

Figure 9: Average values of skin redness.

Skin redness of the test persons increased after SDS exposure.
1% **Ectoin®** **natural** was sufficient to reduce the skin redness by 38% after only 7 days of treatment due to its anti-inflammatory properties.
The efficacy of 1% **Ectoin®** **natural** was equivalent or even better than the results of the treatment with a cream containing 0.25% hydrocortisone.

Reduction of TEWL and skin barrier repair

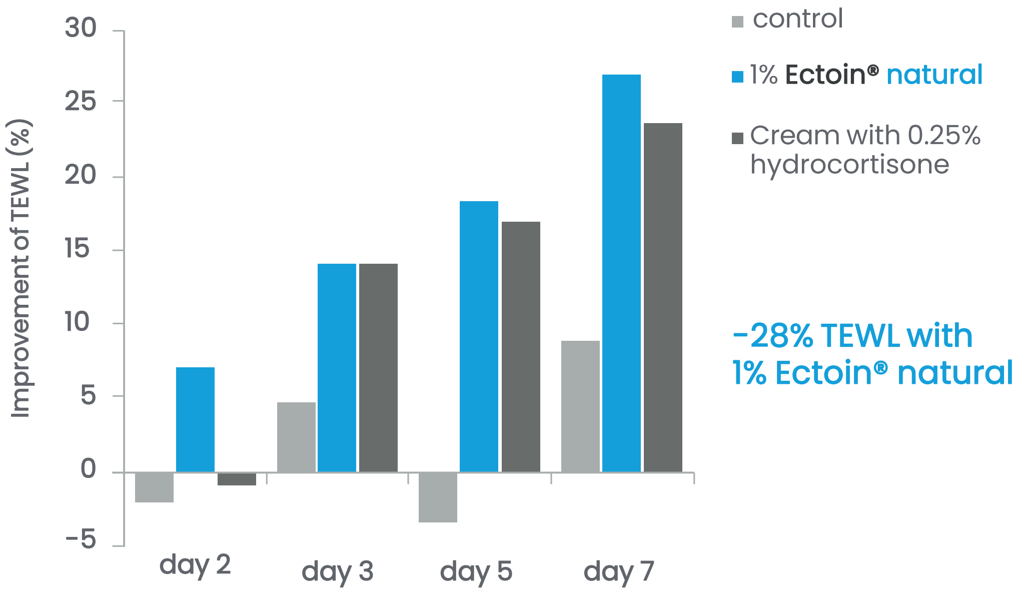


Figure 10: Average values of TEWL.

The TEWL (trans-epidermal water loss) was also measured during the clinical study. TEWL increased after SDS stress.
1% **Ectoin®** **natural** was sufficient to reduce TEWL by 28% after only 7 days of treatment due to its repairing properties.
The efficacy of 1% **Ectoin®** **natural** was equivalent or even better than the results of the treatment with a cream containing 0.25% hydrocortisone.

Tolerance of cosmetic formulations is an ongoing issue - among consumers as well as manufacturers. Skin condition, subjective feeling, and the objective properties of substances play a role in this context.
When it comes to skin irritations, concerned consumers usually try to substitute triggering substances or lower the concentrations. Reducing the concentration of an irritative substance automatically leads to reduced efficacy - a problem with active ingredients such as retinol, which is beneficial as an anti-aging agent but is also highly irritative.[4,5] This problem can be solved by incorporating substances into the formulations that reduce this irritative impact and protect the skin, especially sensitive skin. The sensitivity threshold depends on the concentration of the irritative substance as well as the condition and sensitivity of the skin. In case of highly barrier disordered skin, even substances which are normally harmless may trigger redness, itching, burning sensation, swelling, or irritation.

Ectoin® natural is very well known for its anti-irritative and anti-inflammatory benefits. Its presence allows the use of irritative active ingredients like retinol on sensitive skin conditions, even in high dosages.

SOOTHING AND SKIN BARRIER REPAIR EFFICACY AGAINST RETINOL-INDUCED SKIN IRRITATION

In an *in vivo* study, the skin was treated with retinol in different concentrations to induce skin irritations. According to studies – e.g., Kim *et al.* – retinol and its derivatives may cause severe local irritation manifested as mild erythema and stratum corneum peeling of the skin.[5]
Adding **Ectoin®** natural to a retinol formulation prevents the occurrence of side effects of retinol and allows higher and more effective concentrations of retinol to be used.

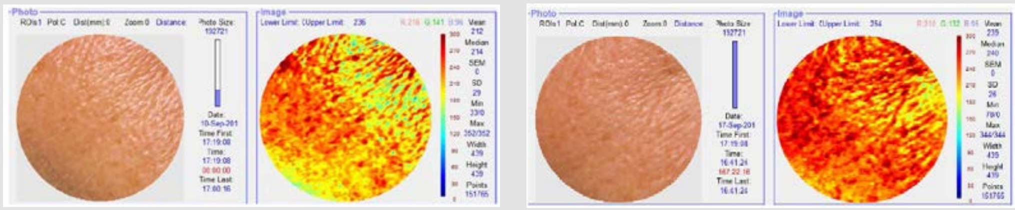
Study protocol

- placebo-controlled *in vivo* study, 23 female volunteers (23-68 years) with sensitive skin
- 2 placebo formulations: 0.5% and 1% retinol
- 2 verum formulations: 0.5% retinol + 1% **Ectoin®** natural; 1% retinol + 1% **Ectoin®** natural
- Application of formulations on the face:
  - week I: 3x/week application of placebo formulations
  - week II: no application of any product (break)
  - week III: 3x/week application of verum formulations (with 1% **Ectoin®** natural)
- measured parameters: soothing effect by measuring red blood concentration in the skin (Tivi 700®) and repairing effect by measuring TEWL (Tewameter®)

SOOTHING EFFICACY *(IN VIVO)*

0.5% Retinol 3x/week

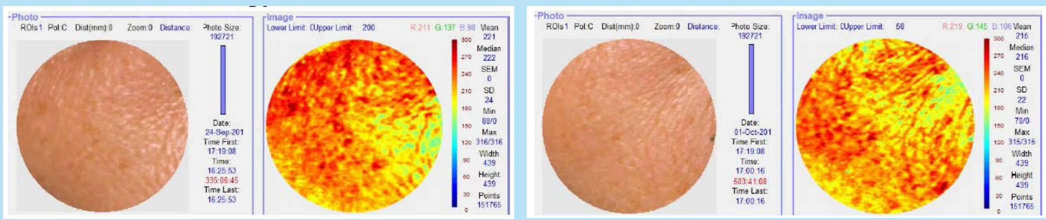
Placebo
(0.5% Retinol)
application
3x in week I



Subject #42 D0 D7

0.5% Retinol + 1% Ectoin® natural 3x/week

Verum
(0.5% Retinol +
1% Ectoin® natural)
application
3x in week III
after 7 days break



Subject #42 D14 D21

Figure 11: In vivo photographic images and Tivi®700 measurements obtained from a representative volunteer show the soothing effect of 1% Ectoin® natural in a 0.5% retinol formulation. (D8-D13: no application)

Results

After three applications of placebo cream (0.5% retinol) during week I, the red blood concentration in the skin increased significantly which resulted in visible skin erythema (skin redness).

After a pause of one week, the verum (0.5% retinol + 1% **Ectoin®** **natural**) was applied three times during week III. 1% **Ectoin®** **natural** reduced the red blood concentration significantly, which resulted in a visible and significant reduction of skin redness (erythema).

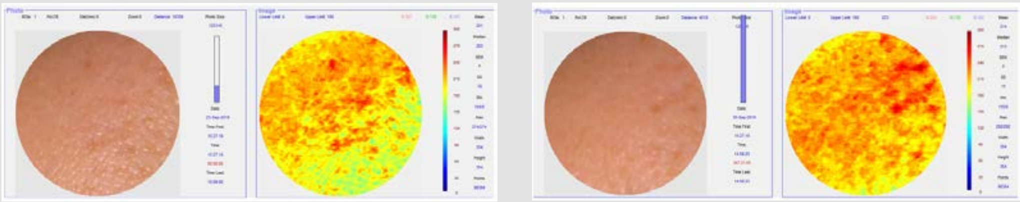
Moreover, the cutaneous microcirculation significantly regains its initial state after treatment with 1% **Ectoin®** **natural**, applied on skin which was irritated with 0.5% retinol formulation.
1% **Ectoin®** **natural** showed a significant soothing and repairing effect on skin irritated with 0.5% retinol.

For skin treated with 0.5% retinol, 1% **Ectoin®** **natural** reduced the skin redness significantly by 8% compared to the placebo treatment.

SOOTHING EFFICACY *(IN VIVO)*

1% Retinol 3x/week

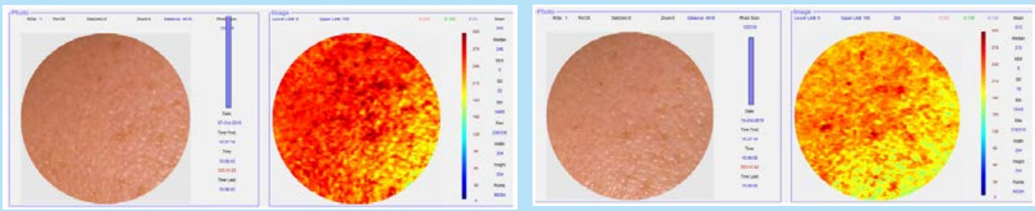
Placebo
(1% Retinol)
application
3x in week I



Subject #55 D0 D7

1% Retinol + 1% Ectoin® natural 3x/week

Verum
(1% Retinol +
1% Ectoin® natural)
application
3x in week III
after 7 days break



Subject #55 D14 D21

Figure 12: In vivo photographic images and Tivi®700 measurements obtained from a representative volunteer show the soothing effect of 1% Ectoin® natural in a 1% retinol formulation. (D8-D13: no application)

Results

After three applications of placebo cream (1% retinol) during week I, the red blood concentration in the skin increased which resulted in visible skin erythema (skin redness).

During the application pause (one week), the irritant impact of the previous application with 1% retinol became visible through a typical increasement in skin redness (delayed reaction). Thus, it can be assumed that the skin was not able to regenerate itself within the seven days pause, the condition even worsened.

After pausing the application for one week, the verum (1% retinol + **Ectoin® natural**) was applied three times during week III. 1% **Ectoin® natural** reduced the red blood concentration, resulting in a visible reduction of skin redness (erythema).

1% **Ectoin® natural** showed a soothing and repairing effect on skin irritated with 1% retinol.

REPAIRING EFFICACY *(IN VIVO)*

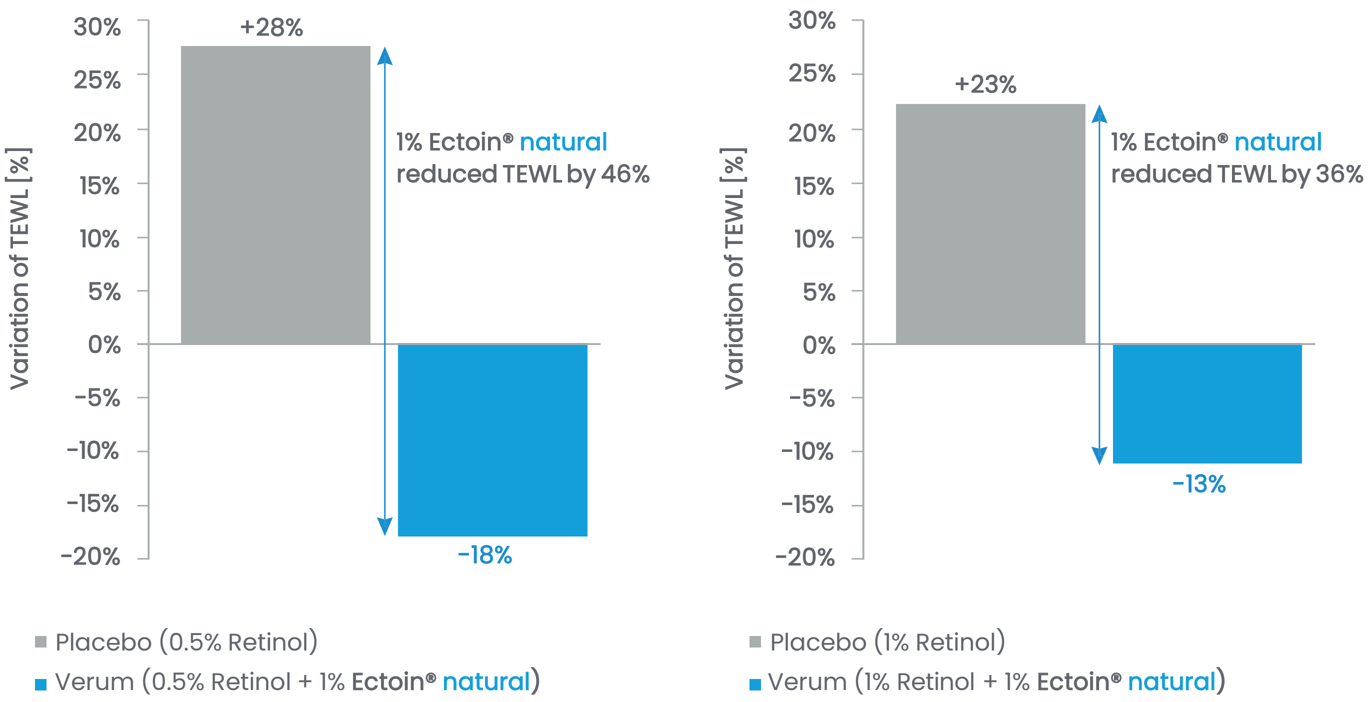


Figure 13: Average values of TEWL of placebo (0.5 and 1% retinol) (grey) and verum (0.5 and 1% retinol + 1% Ectoin® natural) (blue).

The TEWL (trans-epidermal water loss) was also measured during the clinical *in vivo* study.

Results
After three applications during week I of placebo creams (0.5 & 1% retinol), TEWL increased significantly, resulting in cutaneous barrier irritation of the skin.

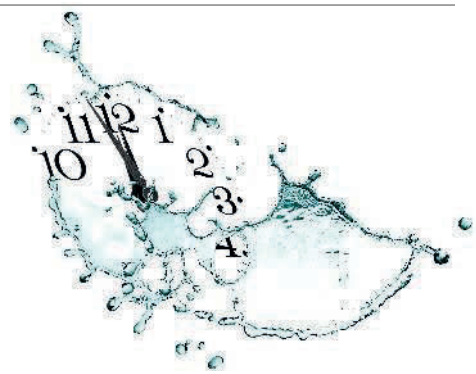
After one week of pause, the two verum formulations (0.5% retinol + 1% **Ectoin® natural**; 1% retinol + 1% **Ectoin® natural**) were applied 3 times during week III.

The verum with 1% **Ectoin® natural** improved the TEWL significantly. Also, the TEWL significantly regains its initial state after the treatment with verum, applied on skin, irritated by 0.5% and 1% retinol formulation (placebo).

1% **Ectoin® natural** showed a significant repairing effect on skin irritated with 0.5% retinol, resulting in a significant reduction of TEWL by 46% compared to placebo treatment.

1% **Ectoin® natural** showed skin barrier repair efficacy by significantly reducing the TEWL by 36% compared to placebo treatment for skin irritated with 1% retinol (placebo).

LONG-LASTING SKIN HYDRATION ("7 DAYS") *(IN VIVO)*



Study protocol:

- double blind, placebo-controlled application test
- 5 female volunteers with normal to dry skin
- cream with 0% (placebo), 0.5% and 1% **Ectoin®** natural, applied 2x/day for 12 days on forearm
- skin hydration was measured from day 8 to day 12
- on day 12, application of cream was stopped for 7 days
- on day 19: detecting the skin hydration again for final result

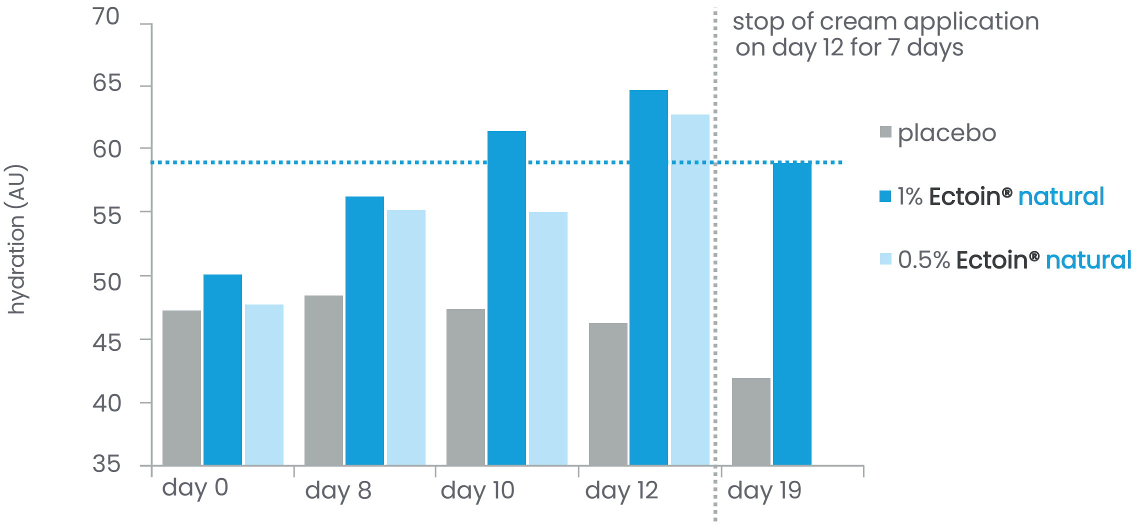


Figure 14: Long-term moisturizing effect with Ectoin® natural. Source: Graf *et al.*, 2008.

Results

During the study, the hydration level in the skin increased up to 200% compared to the placebo-treated skin.

Although the topical application was stopped on day 12 for the following seven days, the hydration status was preserved during this period, underlining a significant long-term moisturizing effect of **Ectoin®** **natural**.[6] The long-term hydration efficacy of **Ectoin®** **natural** is based on its repairing properties and the consequent enhancement and restoration of the skin barrier. After 10 days of application, the increased hydration level remained nearly constant until the end of the study period (day 19).

The anti-pollution efficacy of **Ectoin®** natural has been confirmed by numerous studies (clinical, *in vitro* , and *in vivo*). Until today, it is the only anti-pollution active ingredient which is also approved for the use in medical products and applications including the treatment and prevention of pollution-induced lung diseases (e.g. COPD and asthma). **Ectoin®** natural has been furthermore suggested by independent experts for the use as an anti-pollution active.[5]

CONFIRMED EFFICACY FOR MEDICAL AND COSMETIC APPLICATIONS

For the development of an inhalation solution and other medical devices, several clinical studies were conducted with **Ectoin® natural**.
The clinical studies demonstrated a reduction of inflammations in human lungs of patients with COPD and pollution-induced asthma.[8] Also, the protection of the nose and eye from the penetration of airborne allergens (pollens) was shown.
Please contact us for further information: **cosmetics@bitop.de**

PROTECTION BEYOND PM2.5

Air pollution consists of various components and particles in different sizes. Particulate matter particle sizes are classified into "PM" and so-called "UP" or "UFP" (ultrafine particles; nanosize).

According to the WHO, ultrafine particles constitute over 80% of particulate matter, although their total mass is usually small in comparison to the mass of a few larger particles. The WHO further states, that the highest level of concentration of toxic compounds is related to ultrafine particles (PM0.1 and smaller).

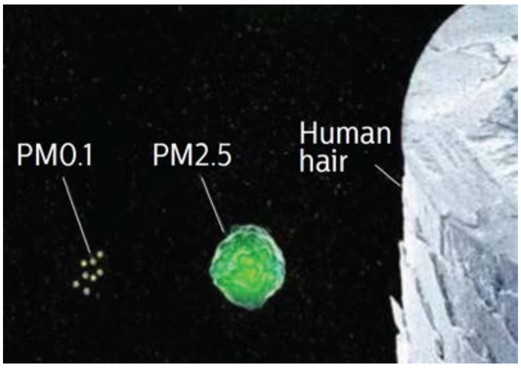


Figure 15: visualization of PM2.5 and PM0.1 particle sizes.

Ectoin® natural is the only anti-pollution active ingredient that protects the skin from PM of all sizes (including PM0.1 - ultrafine particulate matter, also known as UFP) as well as PAHs, heavy metals, nitrogen dioxide, and other toxic air pollution components.

Ectoin® natural offers global and powerful anti-pollution efficacy, repairs environmentally stressed skin and regenerates broken skin barriers.

For comprehensive skin protection against urban stress and prevention of pollution-aging.

PREVENTION OF POLLUTION INDUCED CELL DAMAGE AND SKIN AGING

The *ex vivo* study was carried out by the IUF Leibniz Research Institute for Environmental Medicine (Univ.-Prof. Dr. med. Jean Krutmann), Düsseldorf, Germany.[7] The aim of the study was to show that **Ectoin®** natural protects the skin from pollution-induced damage and aging.

Study protocol

- *ex vivo* study on fresh human epidermal keratinocytes from female Asian and female Caucasian donors
- untreated and pre-treated (24 h) with 2 mM **Ectoin®** natural solution
- stressed with fine and ultrafine carbon black particles and different types of exhaust particles (1.5 μg/cm²)
- determination of POMC, MMP1 and Cyp1A1 expression in untreated and pre-treated keratinocytes

USED POLLUTION PARTICLES TO STRESS THE KERATINOCYTES:

- ultrafine particles (“Printex90”): 0.014 μm particle diameter (nanosize) = PM0.014
- fine particles (“Huber 990”): 0.26 μm particle diameter = PM0.26

Furthermore, two so called “exhaust particles” were used. These are particulate matter (PM) standard reference materials (SRM) for the use in environmental and toxicological methodology and research:

- diesel engine exhaust soot from heavy-duty equipment engines (“SRM 1650”)
- diesel soot from a forklift engine (“SRM 2975”)

Both materials contain PAHs, heavy metals as well as PM10, PM2.5, and PM1.

1. Measured parameter: POMC (proopiomelanocortin) expression

- POMC is a marker for pigmentation and age spot formation.
- POMC peptides (e.g., produced by keratinocytes) are very potent in stimulating melanogenesis in human melanocytes and thus cause pigmentation. The more POMC is measurable in the cell, the more pigmentation and age spot formation is expected on the skin.
- POMC expression in keratinocytes is induced by environmental air pollution components like fine and ultrafine particles (e.g. diesel exhaust) as well as industrial emissions loaded with heavy metals and noxious chemicals.

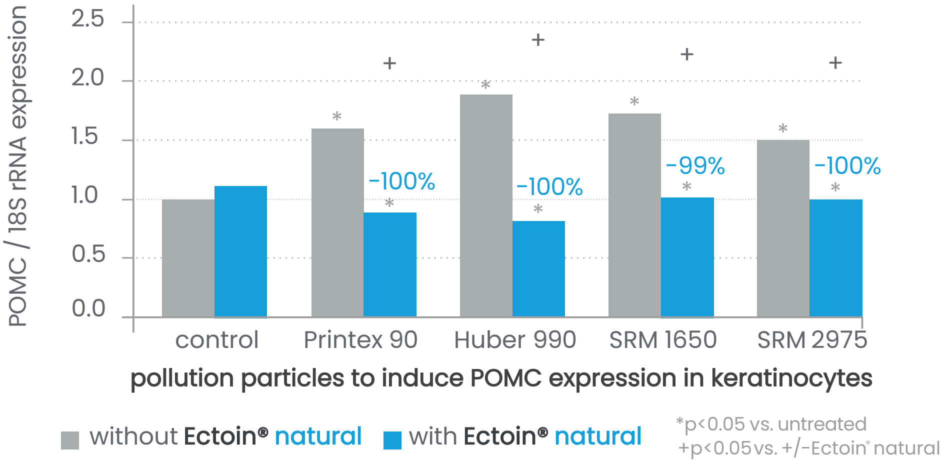


Figure 16: Up-regulation of POMC mRNA in adult primary human keratinocytes by subtoxic amounts of ultrafine (size: 0.014) and fine (size: 0.26 µm) carbon black particles and by two different types of exhaust particulates.

Ultrafine and fine particles, as well as exhaust particulate matter, significantly induce a dramatic up-regulation of POMC in keratinocytes.

Inhibition of pollution-induced POMC expression by Ectoin® natural

used pollution particles to induce gene expression in keratinocytes	POMC up-regulation in keratinocytes treated with Ectoin® natural
Printex90 (ultrafine particles): 0.014 µm particle diameter (nanosize)	0%
Huber990 (fine particles): 0.26 µm particle diameter	0%
SRM1650 (engine exhaust soot): contains PAHs & heavy metals, particle sizes PM2.5, PM10, PM1	1%
SRM2975 (diesel soot): contains PAHs & heavy metals, particle sizes PM2.5, PM10, PM1	0%

Results

The treatment of the skin cells with **Ectoin® natural** completely prevented pollution-induced POMC expression. 100% protection. 0% damage.

Ectoin® natural STOPS pollution-induced up-regulation of POMC in skin cells (Asian and Caucasian) and thus prevents particulate matter-induced skin pigmentation.

2. Measured parameter: MMP1 (matrix metallopeptidase1) expression

- MMP1, also known as collagenase-1, is a marker for wrinkle formation.
- MMPs are enzymes that, when activated (e.g., by pollution particles, UV), control tissue degradation in the dermis. MMPs include collagenase that specifically decomposes particular collagens or other proteins in the extracellular matrix of the dermis. It breaks down the different types of collagen and elastin.
- Environmental air pollution components like fine particles (e.g., diesel exhaust), as well as industrial emissions loaded with heavy metals and noxious chemicals, activate the MMP1 expression in the skin.

Reduction of pollution-induced MMP1 expression by Ectoin® natural

Fine particles as well as engine exhaust particulate matter significantly induce up-regulation of MMP1 in keratinocytes. Keratinocytes derived from Caucasian skin showed higher increase of MMP1 compared to Asian keratinocytes.

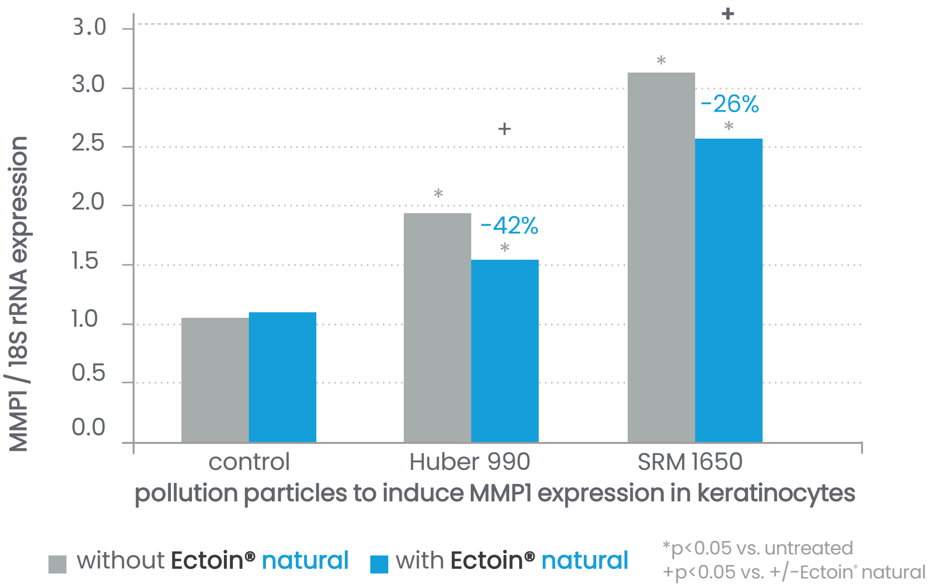


Figure 17: Up-regulation of MMP1 mRNA in adult primary human keratinocytes.

Results
Treatment of the skin cells with **Ectoin® natural** significantly reduced MMP1 expression by up to 42%.
Consequently, pollution-induced wrinkle formation can be reduced by **Ectoin® natural** in Asian and Caucasian skin.

3. Measured parameter: Cyp1A1 expression

- Cyp1A1, also known as AHH (aryl hydrocarbon hydroxylase), is a marker for PM-induced cell damage.
- Cyp1A1 is involved in the activation of polycyclic aromatic hydrocarbons (PAHs). PAHs are air-pollution components (environmental chemicals). PAHs acquire carcinogenicity after they have been activated by Cyp1A1 enzymes to highly reactive metabolites capable of attacking cellular DNA.
- Environmental air pollution components like fine and ultrafine particles, (e.g., diesel exhaust) as well as industrial emissions loaded with heavy metals and noxious chemicals, induce the Cyp1A1 expression in skin cells.

Reduction of pollution -induced CYP1A1 expression by Ectoin® natural

Ultrafine (nanosize), fine particles as well as different kinds of exhaust particulate matter significantly induce the up-regulation of Cyp1A1 in primary human keratinocytes.

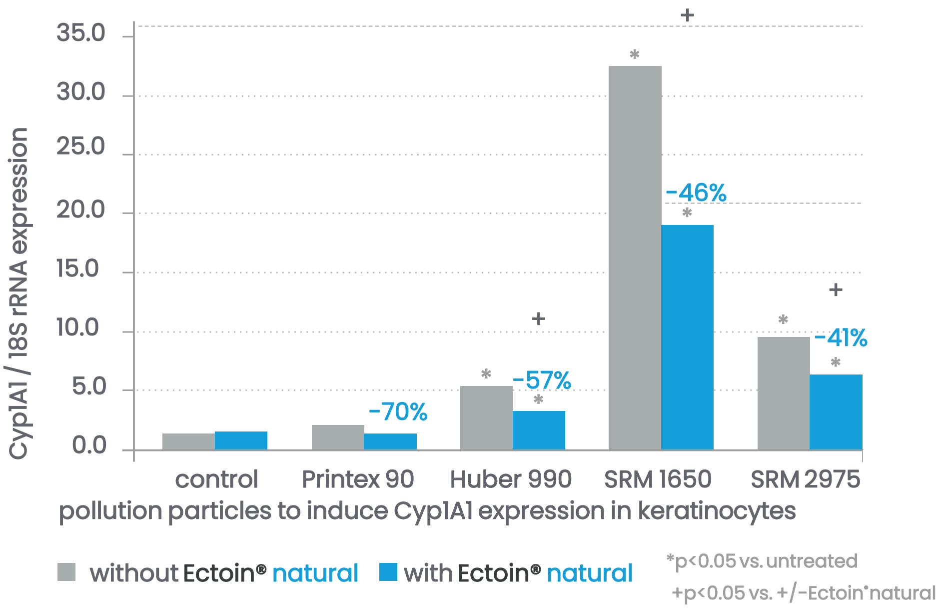


Figure 18: Up-regulation of Cyp1A1 mRNA in adult primary human keratinocytes.

Results
The treatment of skin cells with **Ectoin®** **natural** resulted in a significant reduction of the pollution-induced Cyp1A1 expression by up to 57%, demonstrating the strong cell-protection effect of **Ectoin®** **natural** against pollution particles.

PROTECTION FROM POLLUTION-INDUCED SKIN DAMAGE *(IN VIVO)*

The purpose of the study was to confirm the anti-pollution efficacy of **Ectoin®** natural in a specific, objective and reliable *in vivo* setting.
This *in vivo* study was carried out by a specialized dermatological center in Germany. The study design is the most innovative, controlled, and standardized *in vivo* pollution test method currently available.

Study protocol

- in vivo study (placebo-controlled, randomized, double blind) on volar forearm of six volunteers
- 2x/day application of placebo or 1% **Ectoin® natural** (verum) for 5 days
- on day 5: skin was stressed with cigarette smoke as pollutant for 15 min using a standardized pollution chamber system
- measured parameter: concentration of Malondialdehyde (MDA) in untreated unstressed skin (neg. control), in untreated/stressed skin (pos. control), in placebo-treated stressed skin and in **Ectoin® natural** treated/stressed skin.

MDA results from lipid peroxidation of polyunsaturated fatty acids in the skin and is one of the reactive electrophile species that cause toxic stress in skin cells. Therefore, it can be used as a marker for air pollution-induced skin damage.

Cigarette smoke application on the volar forearm after 5 days of treatment with 1% **Ectoin®** natural

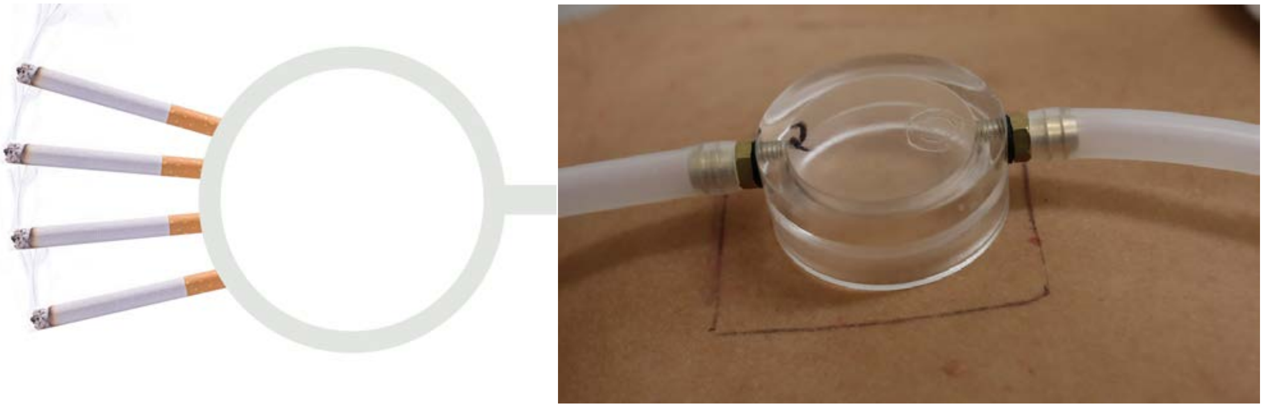


Figure 19: *In vivo* anti-pollution study design.

Cigarette smoke contains all common classes of air pollutants. It is composed of more than 4300 different chemicals including high levels of free radicals and up to 500 ppm nitric oxide (NO), which slowly undergoes oxidation to nitrogen dioxide (NO_2). Per puff, more than 1014 low molecular-weight carbon- and oxygen-centered radicals can be measured. Therefore, cigarette smoke is an excellent pollutant for *in vivo* anti-pollution testing.

RESULTS: Reduction of pollution induced skin damage

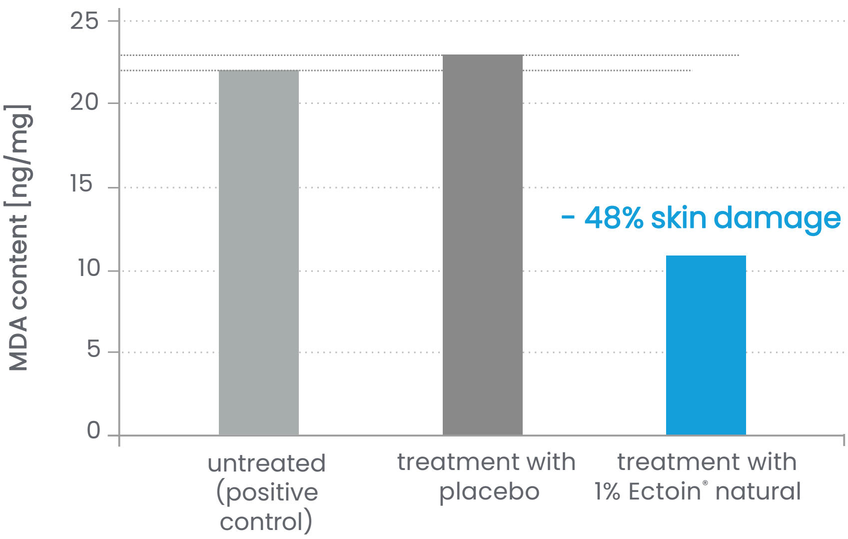


Figure 20: Reduction of pollution-induced skin damage.

After only 5 days of application with 1% **Ectoin®** **natural** the pollution-induced MDA content in the skin was reduced by 48% compared to the placebo-treated areas.
In the same study set-up, a strong anti-oxidant was capable to reduce the pollution-induced MDA content in the skin by only 27% (data not shown).

Results
Ectoin® **natural** shows global anti-pollution efficacy.
After only 5 days of application, the pollution-induced lipid peroxidation in the skin was remarkably reduced by 48%.
Ectoin® **natural** is capable of shielding the skin from the whole spectrum of air pollution components including heavy metals and PAHs as well as all particle sizes (including PM0.1 and smaller).
Ectoin® **natural** provides complete and instant protection from pollution-induced skin damage and aging, including skin pigmentation.
Pollution–stressed and irritated skin is soothed and comprised skin barrier is repaired.

BLUE LIGHT PROTECTION

“Visible light” is a part of the sunlight spectrum and is visible to the human eye. It constitutes around 39% of the solar radiation (wavelength: 390-700 nm) and includes the high energy visible (HEV) light, better known as blue light (400–525 nm). In the past years, numerous studies on the effects of blue light on the skin were performed, showing that blue light induces oxidative stress responses and DNA damage in the epidermis and dermis.[2]

H2AX is a part of the H2A histone family. Histones are important proteins in the human cells acting as spools that DNA winds around to form structural units, known as nucleosomes.[11] If H2AX becomes phosphorylated due to DNA double-strand breaks, it is called γH2AX. Therefore, the phosphorylated form (γH2AX) is used as an indicator of DNA damage.

Ectoin® natural is capable to protect the cells and prevent DNA damage in the skin induced by visible/blue light. Due to the protection efficacy of **Ectoin®** natural, the cells of the epidermis show improved cell functions like a faster supply of protection proteins and self-defense reactions.[12]

In an in *vitro* study, NHEK were exposed to blue light (220 J/cm^2) daily for three days to show the protective efficacy of **Ectoin®** natural.

Study protocol

- normal human epidermal keratinocytes (NHEK, 23 years old, Caucasian woman)
- 0.5% **Ectoin®** natural in growth medium (Promocell) vs. placebo; incubation for 24 h before and throughout blue light exposure
- blue light exposure: 220 J/cm^2 (470 nm), 1 h 50 min daily for 3 days
- measured parameters: immunolabeling of γH2AX

PROTECTION EFFICACY *(IN VITRO)*

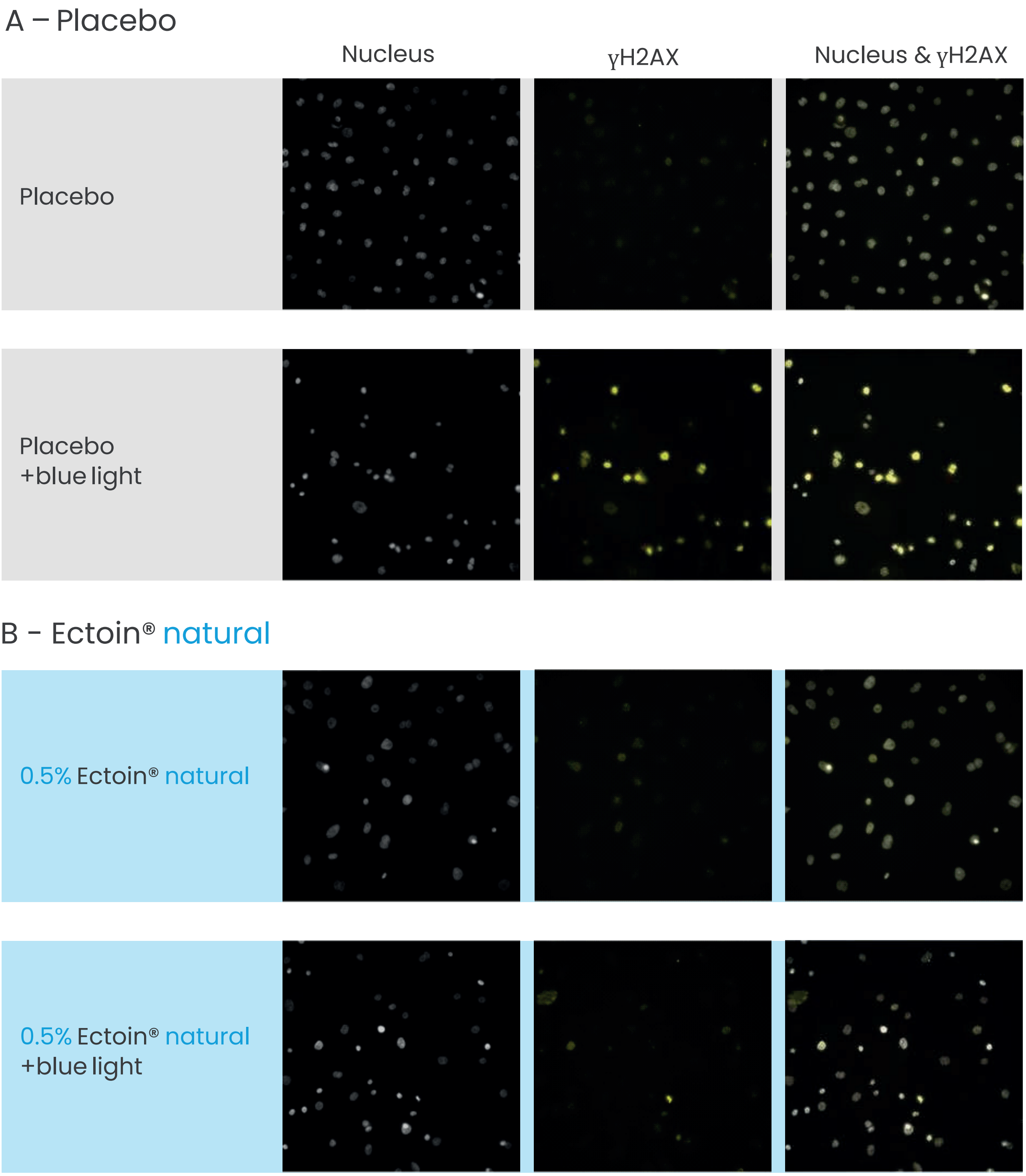


Figure 21: Fluorescence images of NHEK cells (γH2AX stained in yellow, nuclei stained in grey) (A) placebo before and after blue light irradiation (220 J/cm², 1 h 50 min daily for 3 days); (B) 0.5% Ectoin® natural before and after blue light irradiation (220 J/cm², 1 h 50 min daily for 3 days)

PROTECTION EFFICACY *(IN VITRO)*

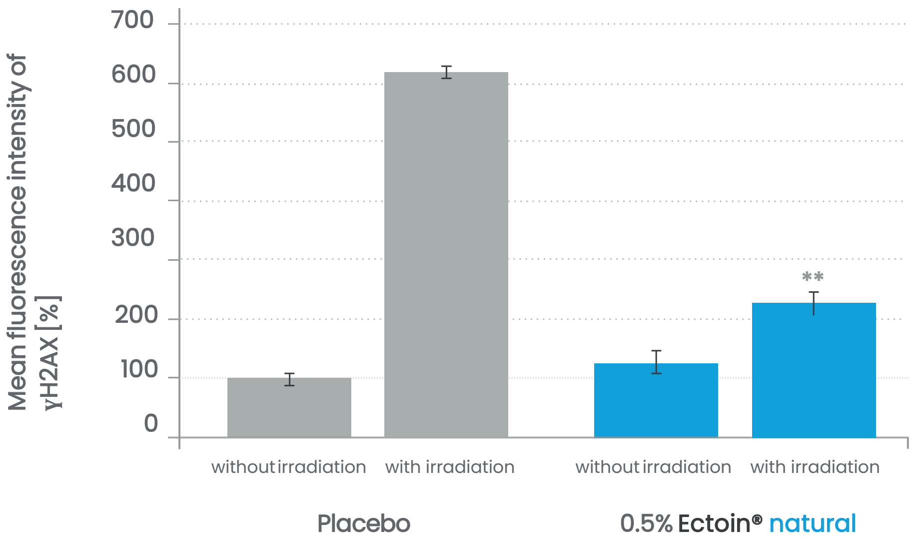


Figure 22: Average fluorescence intensity of γH2AX [%] in NHEK before and after blue light irradiation (220 J/cm², 1 h 50 min daily for 3 days) of placebo (grey) and 0.5% Ectoin® natural (blue).

Results
After daily exposure (1 h 50 min) to blue light for three days, a six-fold increase in γH2AX content was observed in the placebo control. This significantly higher amount of γH2AX indicates DNA damage by blue light in the keratinocytes.
0.5% **Ectoin® natural** significantly reduced the phosphorylation of H2AX and thus prevents blue light-induced DNA damage.

VISIBLE/BLUE LIGHT PROTECTION *(EX VIVO)*

Several studies showed that **Ectoin®** natural is able to prevent oxidative damage in skin induced by visible/blue light. Due to its comprehensive protection efficacy against the whole solar spectrum on cellular level, **Ectoin®** natural completes sun protection formulations.

Study protocol

- human living skin explants (18 years old Caucasian woman)
- topical treatment (2 µl/explant) with placebo and 1% **Ectoin®** natural for 4 days and untreated control
- exposure to visible light (65 J/cm^2) on day 4; sampling of skin explants on day 5
- evaluated parameters: immunostaining of Nrf2 and MC1R

RESULTS: Ectoin® natural reduces blue-light induced oxidative stress

Nrf2 is a key transcription factor in the cellular response to oxidative stress. It protects skin from light-induced stress. Nrf2 content in the cell increases automatically when the skin is exposed to stress factors like visible/blue light.

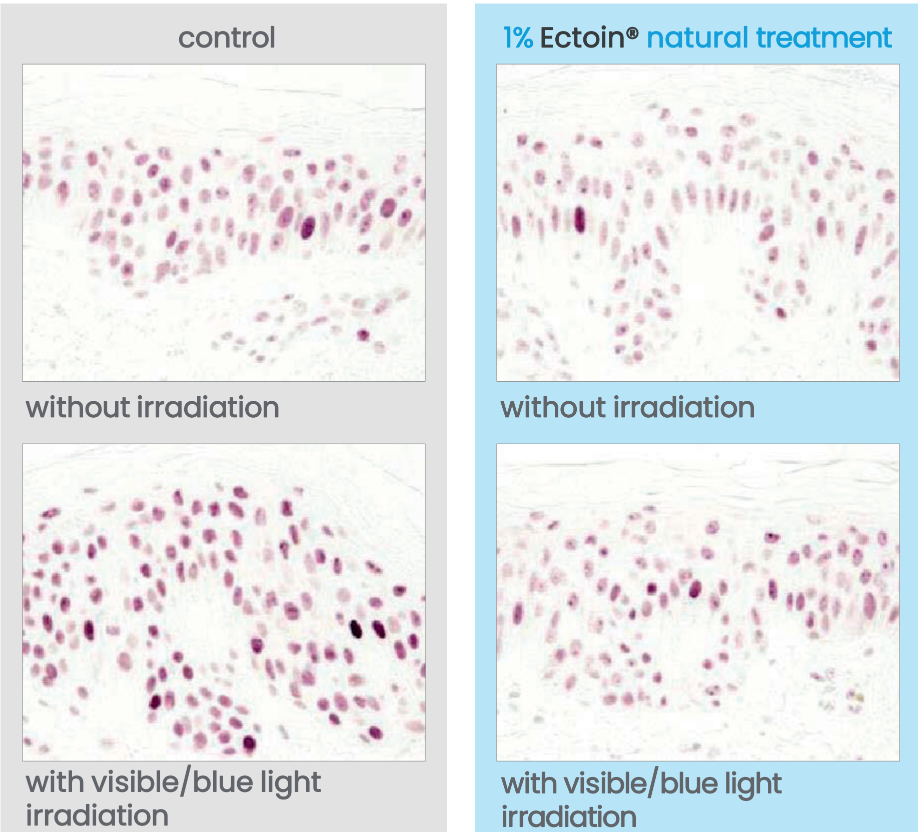


Figure 23: Nrf2 immunostaining of the skin explants. Nrf2 content is dyed in red (red dots). The darker the color, the higher the Nrf2 content in the nucleus.

Results
Pre-treatment of the skin explants with 1% **Ectoin®** **natural** resulted in a reduction of visible/blue light-induced skin damage.
The skin explants treated with **Ectoin®** **natural** and irradiated with visible/blue light showed significantly lower Nrf2 content compared to control and placebo treatment. **Ectoin®** **natural** was able to reduce the oxidative stress response of the cell to visible/blue light irradiation.

RESULTS: reduction of blue-light induced pigmentation by Ectoin® natural

MC1R (melanocortin 1 receptor) is a marker for pigmentation.
Visible light irradiation of the skin explants leads to an over-expression of MC1R, initiating a pigmentation response in the skin.

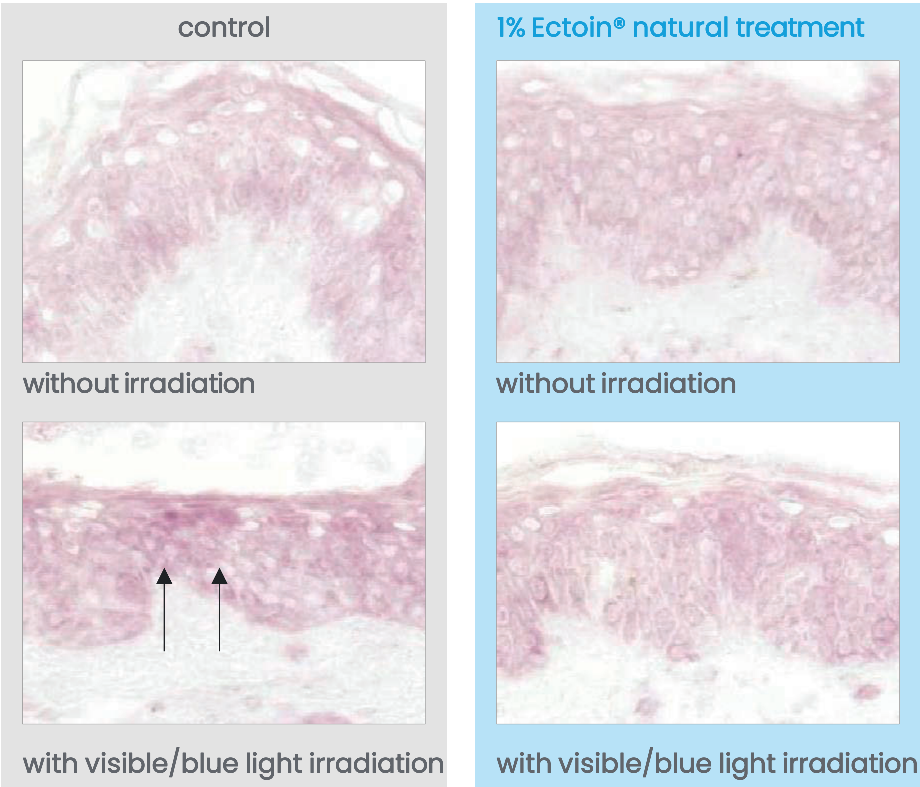


Figure 24: MC1R immunostaining of the skin explants. MC1R is dyed in red. The darker the color, the higher the MC1R content.

Results
Pre-treatment of the skin explants with 1% **Ectoin®** **natural** resulted in a reduction of visible/blue light-induced melanogenesis.
The skin explants treated with **Ectoin®** **natural** showed significantly lower MC1R content compared to control and placebo treatment. **Ectoin®** **natural** is able to reduce visible/blue light-induced skin pigmentation.

BRIGHTENING & LIGHTENING EFFICACY *(IN VIVO)*

Due to environmental factors such as solar radiation, blue light, and air pollution, as well as oxidative stress, the skin can experience hyperpigmentation – unwanted freckles and pigment spots increasingly arise.

The aim of the *in vivo* study was to show the brightening and skin lightening efficacy of **Ectoin® natural**.

Study protocol:

- *in vivo* study with 22 female Asian test persons (36-60 years old) with phototype III and IV
- subjects presenting at least one pigment spot with a diameter ≥ 3 mm on the face
- application of cream containing 1% **Ectoin® natural** on face twice daily for 84 days.
- on day 0, 28, 56, and 84: measurement with Spectrocolorimeter®
- on day 84 subjective evaluation
- measured parameter: depigmenting effect, reduction of the different pigmentations, subjective questionnaires

Results: Depigmenting effect and skin lightening

Table 3: Parameters for depigmenting effect.

Parameter	Change in % after 56 days	Change in % after 84 days
Skin lightness	+ 2%	+ 3%
Skin pigmentation	- 25%	- 28%

Results
Skin lightness significantly increased by up to 3%, this effect was observed on 100% of the test persons.
Skin pigmentation decreased significantly by up to 28%.

Results: Reduction of pigmentation

Characterized by a significant decrease of the variation between spot and normal skin

Table 4: Parameters for the reduction of different pigmentations.

Parameter	Change in % after 56 days	Change in % after 84 days
variation between spot and normal skin	**- 11%**	**- 22%**

Results
Variation between pigmented skin and normal skin significantly decreased by up to 22%.
Ectoin® natural can reduce the difference of pigmentation.

Results of the subjective questionnaires

Table 5: Product efficacy after 84 days of use.

Statement	% of positive answers
The size of spots is smaller	96%
The color of spots is attenuated (less dark)	100%
The number of spots is decreased	91%
The skin is lighter	100%
The skin is radiant	100%
The skin appears smoother	100%
The product leaves a uniform complexion	91%
The product leaves a softer skin	100%
The product leaves a hydrated skin	100%

Results
The subjective questionnaire shows a satisfaction by the large majority of the subjects for the properties and efficacy of **Ectoin® natural.**
100% of the test persons stated that the skin was lighter after 84 days of 1% **Ectoin® natural** application.

INFLUENCE ON THE HUMAN MICROBIOME

The human skin is home to millions of microorganisms, such as bacteria, viruses, or fungi that compose the individual skin microbiota. These microscopic organisms bear an important role in maintaining the health of our skin by building a protective layer on the skin against invading pathogens.[13] Furthermore, they are essential for the education of our immune system. A healthy microbiome supports the skin to stay in a good condition, due to pH stabilization and the production of beneficial nutrients. Topic products can affect the microbiome positively by maintaining ("microbiome-friendly") or even promoting the natural growth of microorganisms (prebiotic).

An *in vitro* study was carried out to evaluate the influence of **Ectoin®** natural on four bacterial strains that represent a large percentage of the human skin microbiome.

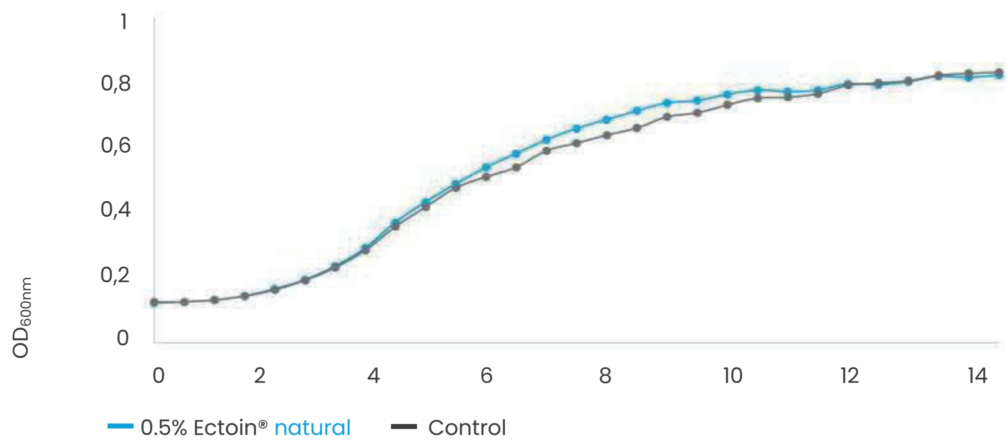
Study protocol

- mix of 4 bacterial strains (*Staphylococcus aureus* ATCC® 6538™, *Staphylococcus epidermidis* ATCC14990™, *Cutibacterium acnes* ATCC® 6919™, *Corynebacterium striatum* ATCC® 6940™)
- incubation in growth medium (EpiLife™) with 0.5% & 1% **Ectoin®** natural, control (placebo) for 48 h
- measured parameter: influence on the microbiome by measuring the growth of the bacterial strains; evaluation of bacterial viability by performing a XTT assay

MICROBIOME FRIENDLY

Growth curve of bacterial mix

A



B

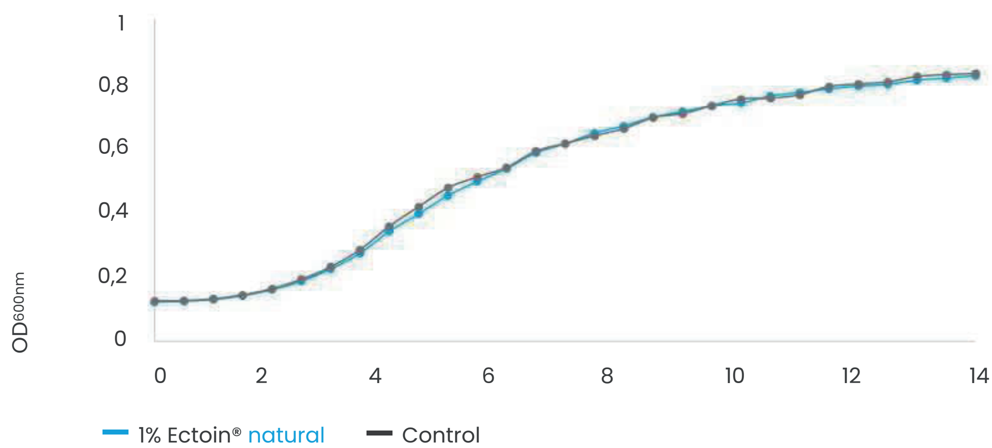


Figure 25: growth curve of bacterial mix of (*Staphylococcus aureus* ATCC®6538™, *Staphylococcus epidermidis* ATCC®14990™, *Cutibacterium acnes* ATCC®6919™, *Corynebacterium striatum* ATCC®6940™) incubated with (A) 0.5% Ectoin® natural (blue) and placebo (grey), (B) 1% Ectoin® natural (blue) and placebo (grey) ; growth from t0 to t14h

Results

0.5% & 1% **Ectoin® natural** had no influence on the bacterial growth. A 100% viability of the bacterial mix (microbiome) could be observed after 48 h of incubation.
Ectoin® natural is a microbiome-friendly active ingredient that has no negative impact on the natural human microbiome.

UVA AND VISIBLE LIGHT PROTECTION

Most commercialized sunscreens show effectiveness against UVB and UVA but do not provide significant protection against visible light. The photoprotective properties of **Ectoin®** natural against the combination of visible light and UVA-radiation as well as visible light only were tested on human dermal keratinocytes.[9]

Study protocol:

- pre-treatment of keratinocytes with **Ectoin®** natural (0.025, 0.05 and 0.1 mM)
- keratinocytes were irradiated with combined light (UVA/visible) and visible light (dose of 15 J/cm²)
- after irradiation, cells were submitted to comet assay for efficacy measurement of photo protection

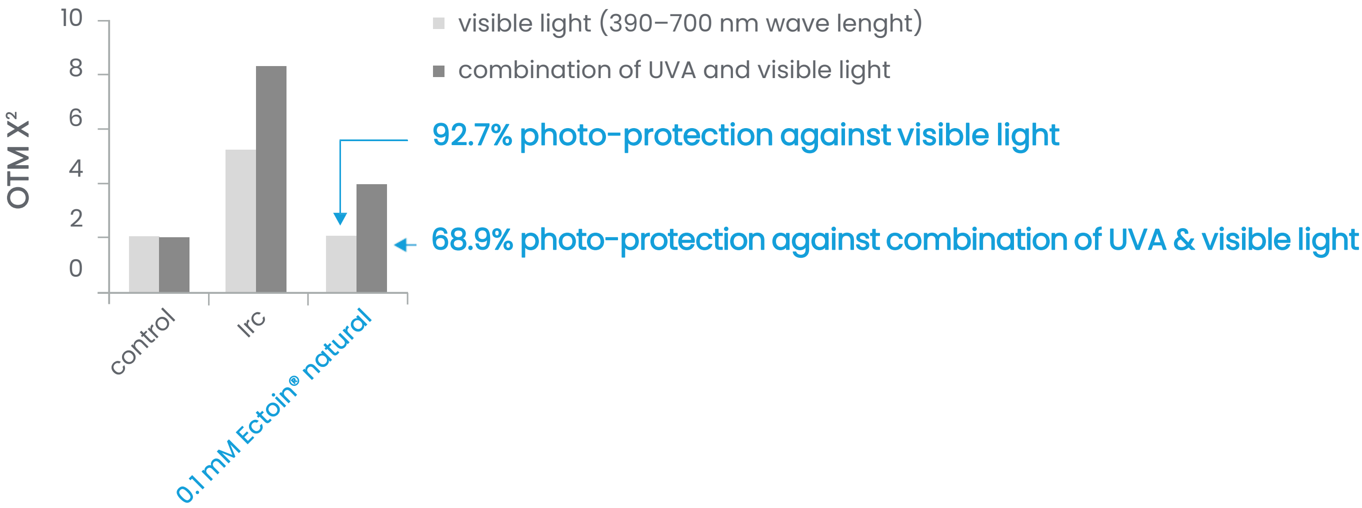


Figure 26: Photo protection (UVA and visible light) of Ectoin® natural on human keratinocytes measured by the alkaline comet assay. Control: non-irradiated cells; Irc: untreated irradiated cells. Source: Botta *et al*, 2008.

Results
All tested concentrations of **Ectoin®** **natural** showed a highly significant level of photoprotection against the combination of UVA/visible light and the irradiation with visible light on cellular level.
Ectoin® **natural** completes sunscreen and day care formulations for full spectrum protection against UVA-radiation and visible light (including blue light) damage. Level of photoprotection: 92.7% (visible light) and 68.9% (UVA/visible).

The protective properties of **Ectoin®** natural regarding the whole solar spectrum were demonstrated in various studies.

Ectoin® natural protects the epidermal self-defense system – the Langerhans cells – from damage by UV radiation *(in vivo)*. It also prevents the formation of sunburn cells developed as a result of the human skin being exposed to UVB radiation.

Ectoin® natural does not act like a filter. It does not absorb the UV radiation but protects the skin by surrounding the skin cells and all other components of our skin with protection shells. This protection mechanism is even capable to shield the skin from the destructive effects of damaging UVA, UVB, IR, and visible light irradiation on cellular level.

UVB-PROTECTION: PREVENTION OF SUNBURN CELLS (SKIN MODEL)

The existence of sunburn cells in the epidermis is an indicator of UV-induced skin damage. Sunburn cells are keratinocytes undergoing apoptosis (controlled cell death) after they have received a UV dose that severely damaged their DNA or other chromophores and repair was no longer possible.

Study protocol:

- organotypic skin model (Skinethic®)
- 24 h pre-treatment with 4% **Ectoin®** natural or placebo
- UVB-irradiation of organotypic skin
- histological proof of sunburn cells

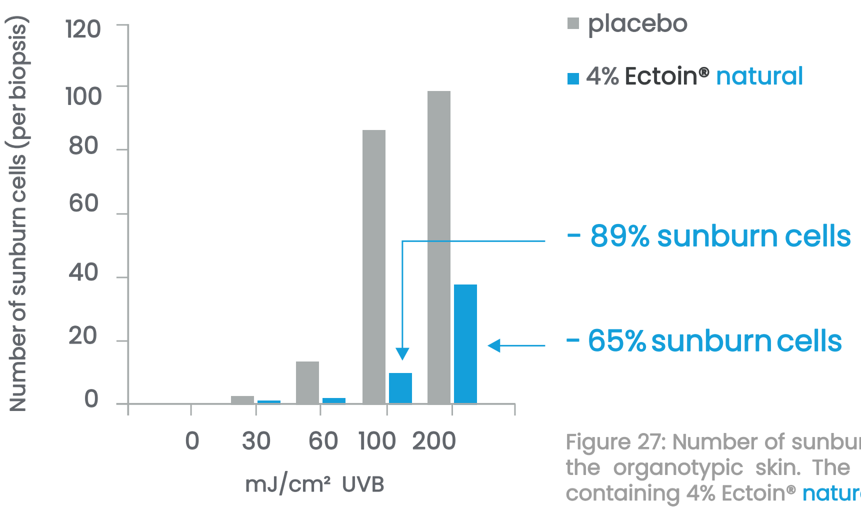


Figure 27: Number of sunburn cells per biopsis after UVB-irradiation of the organotypic skin. The skin was pre-treated with a formulation containing 4% Ectoin® natural or placebo. Bünger *et al.*, 2001.

Results

Ectoin® natural protects the skin against harmful effects and damages induced by UVB radiation. It significantly prevents the formation of sunburn cells due to damaging UVB exposure.[2]

The pre-treatment of the skin model with **Ectoin®** natural reduced the number of sunburn cells by up to 89% compared to placebo.

UV radiation constitutes only 7% of the solar energy that reaches the skin. 39% lies in the visible light spectrum, but the most considerable fraction of solar energy - 54% - consists of infrared radiation.

IR radiation is divided into IR-A, IR-B, and IR-C. Of these, only the IR-A fraction (30% of the IR-radiation that reaches the human body) penetrates deeply into the skin. 65% of this amount reaches the dermis, meaning that IR-A is able to affect the dermis.

Heat is a form of energy that is transmitted by IR radiation, resulting in raised skin temperature. Human skin is exposed daily to natural sunlight. It was found that the temperature of human skin can increase to more than 40 °C under IR-irradiation due to the conversion of IR into heat. There is clinical evidence indicating that chronic heat exposure of human skin is causing alterations.[10]

The latest scientific findings showing the damaging effects of IR on human skin touch on aspects of both prevention and therapy. Classic sun protection strategies which do not include IR-A protection should be reconsidered.

IR-A / HEAT PROTECTION – BOOST OF SKIN´S SELF-DEFENSE

Human skin cells produce heat-shock proteins (HSP) when exposed to external stress like solar radiation or other physical and chemical stress factors. Heat-shock proteins recognize, stabilize and metabolize damaged proteins in the skin. They play an important role in the skin's self-defense system.

Study protocol:

- human keratinocytes pre-treated with 1% **Ectoin®** natural
- expose to heat-shock for 60 min
- measured parameters: stress proteins (HSP 72/73) marked with fluorescent antibodies

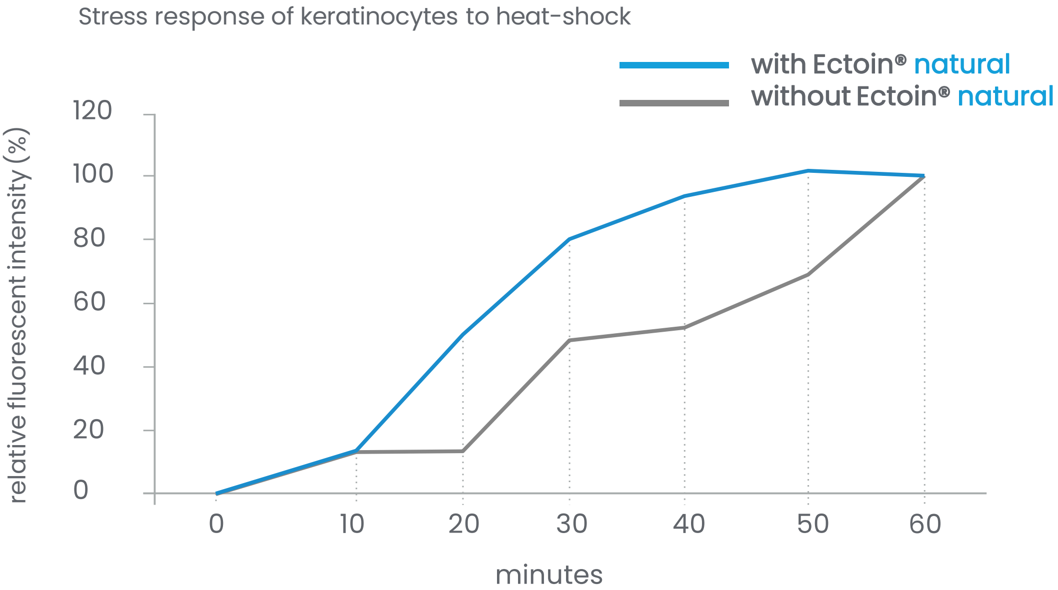
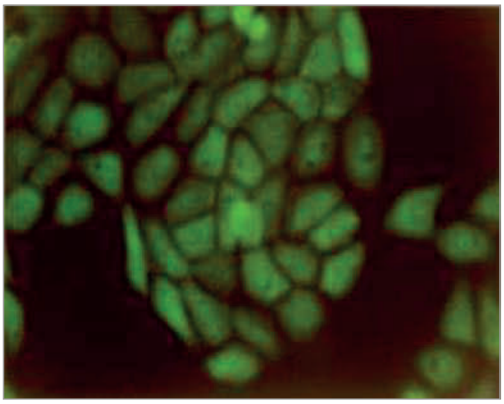
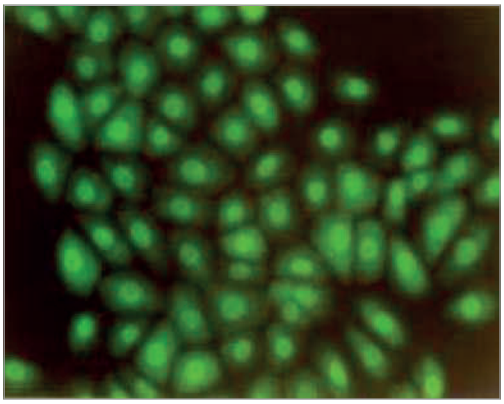


Figure 28: Microscopic analysis of stress response (HSP 72/73 expression) in untreated (grey) and in Ectoin® natural treated (blue) keratinocytes. Source: Bünger and Beyer, 2001.

Stress response of keratinocytes (formation of heat-shock proteins) 30 minutes after heat-shock



without **Ectoin®** natural



pre-treatment with 1% **Ectoin®** natural

Figure 29: Microscopic images of fluorescent HSP 72/73 in the keratinocytes with and without Ectoin® natural pre-treatment.

Results
The pre-treatment of keratinocytes with **Ectoin®** **natural** resulted in a more rapid supply of heat-shock proteins. **Ectoin®** **natural** treated skin cells reacted 2 to 3 times faster to heat stress than untreated cells.[2]
Ectoin® **natural** boosts the supply of heat-shock proteins (HSP) in skin cells. Further *in vitro* studies on living human cells demonstrated the protection benefits of **Ectoin®** **natural** against excessive IR-radiation.[14, 15]
Additionally, **Ectoin®** **natural** shows thermal protection benefits by protecting enzymes like Lactate Dehydrogenase (LDH) and Phosphofructokinase (PFK) from degeneration during heat exposure, showing thermal protection benefits.

INHIBITION OF UVA-INDUCED SKIN DAMAGE

UVA-radiation is the major component of the UV solar spectrum that reaches the skin. Among other things, UVA-radiation initiates the gene regulatory mechanisms in human keratinocytes that are responsible for the damage caused by such radiation. It has been shown that the formation of singlet oxygen in the membrane of UVA-radiated keratinocytes releases ceramides which are second messengers and initiate a UVA-induced activation of the transcription factor AP-2 and expression of pro-inflammatory genes like ICAM-1. In addition to UVA-induced ICAM-1 expression, matrix metalloproteinases are activated, which cause wrinkles by degradation of the extracellular matrix proteins (e.g., elastin and collagen).[18, 19, 20]

Study protocol:

- human dermal keratinocytes, 24 h pre-incubation of keratinocytes with 1 mM **Ectoin®** natural
- irradiation with 30 J/cm² UVA for 1 h
- determination of second messenger (ceramides) release, AP-2 activation and ICAM -1 expression

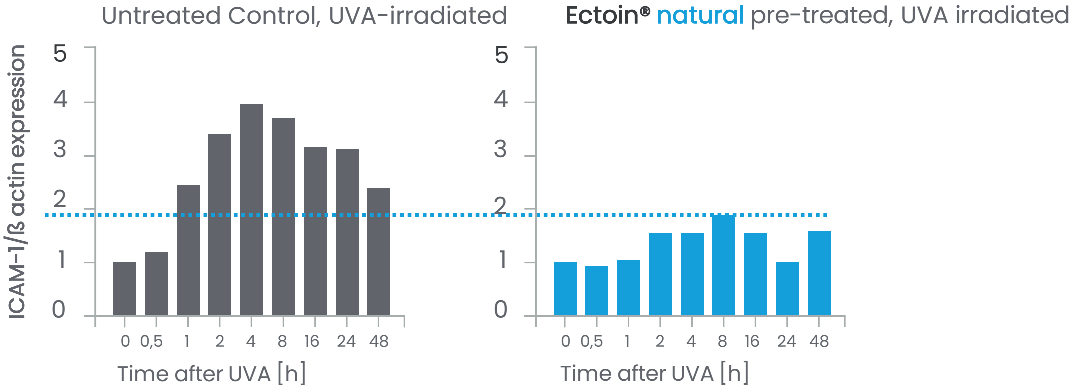


Figure 30: UVA-induced ICAM-1 mRNA expression in comparison with the inhibition of these expressions by Ectoin® natural pre-treatment.

Inhibition of UVA-induced second messenger release with **Ectoin®** natural

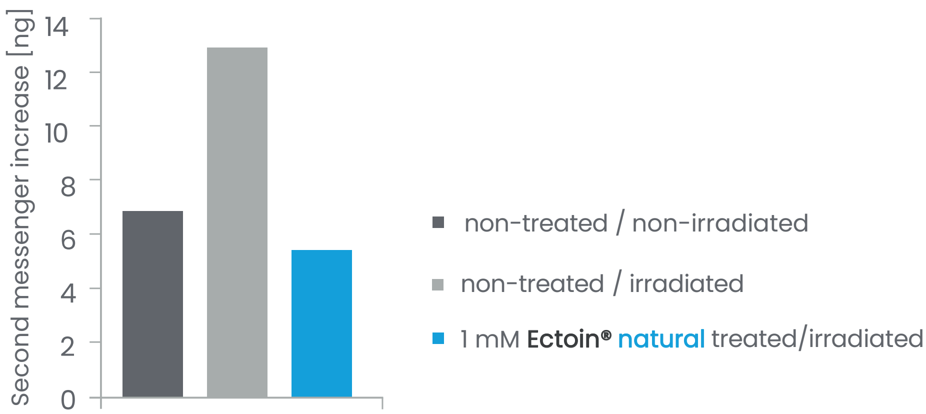


Figure 31: UVA-induced second messenger (ceramides) release in untreated and pre-treated cells with Ectoin® natural.

Results
Ectoin® **natural** completely inhibits the UVA-induced second messenger release, the consequent activation of AP-2, and expression of ICAM-1 in human keratinocytes.[18,19] The measured parameters show that **Ectoin®** **natural** prevents the damaging cascade inside UVA stressed cells.[18] **Ectoin®** **natural** protects skin cells from UVA-induced photoaging on cellular level.

UVA-stress model the pathway to premature photo aging

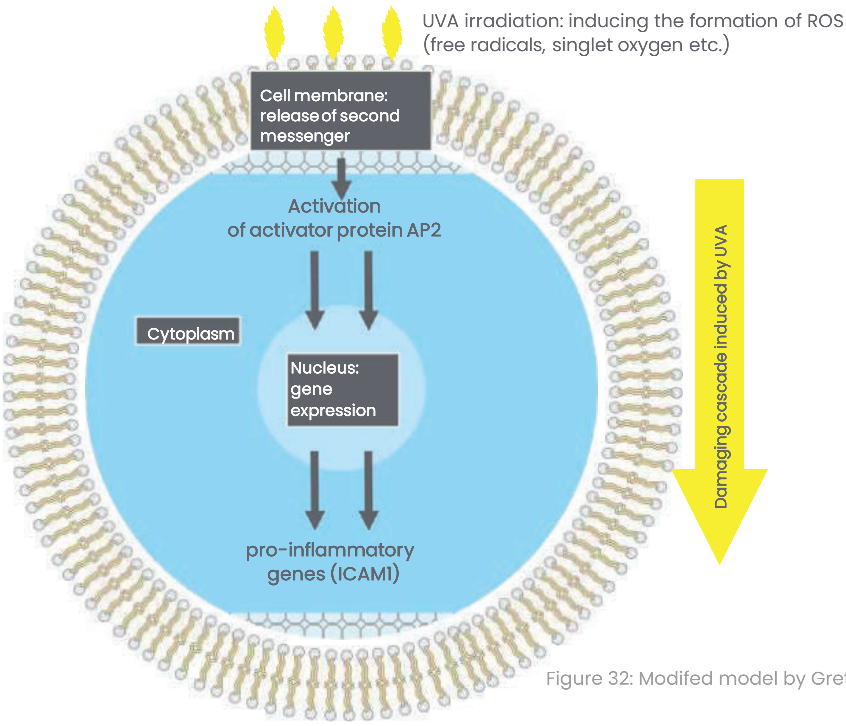


Figure 32: Modifed model by Grether-Beck *et al.* 2005.

BLOCK OF UVA-INDUCED DNA DAMAGE

Repetitive UVA radiation is known to cause time-and dose-dependent mutations and damage of mitochondrial DNA in skin cells, which consequently leads to premature photoaging.

Study protocol:

- human dermal fibroblasts were pre-treated with 1 mM **Ectoin®** natural over 24 h
- fibroblasts were exposed 3x/day to UVA-radiation (8 J/cm²) on four consecutive days over three weeks
- the mutations of the mitochondrial DNA were measured with polymerase chain reaction

Results
Pre-treatment with **Ectoin®** **natural** was sufficient to completely inhibit long-time UVA-radiation induced mitochondrial DNA mutation.[2,18]
The cells showed no degeneration or damage at all.

IMPROVEMENT OF SCALP CONDITION *(IN VIVO)*

Study protocol:

- 30 test persons (18-65 years old) with dandruff, itching, redness or dryness regarding the scalp
- application of scalp tonic with 1.3% **Ectoin®** natural, 2x/day for 4 weeks
- evaluated parameters: skin dryness, skin redness, skin scaling (dandruff) and skin irritation

Improvement of the scalp parameters after 4 weeks of 2x daily application with 1.3% Ectoin® natural solution:

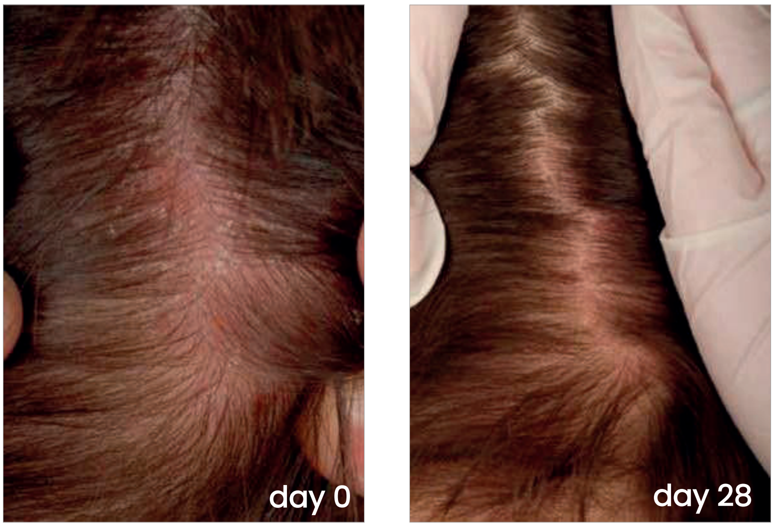


Figure 33: *In vivo* photographic images obtained from a representative volunteer show the improvement in scalp condition achieved after a 4-weeks treatment with 1.3% Ectoin® natural.

Table 6: Percentage changes of means from day 0 to 4 weeks surface.

Parameter	average change in %	maximal change in %
scalp dryness	- 27.0%	- 55.5%
scalp redness	- 38.1%	- 100%
dandruff	- 33.3%	- 59.4%
scalp irritation	- 33.6%	- 65.8%

Results

Ectoin® natural leads to a significant improvement of scalp parameters like dryness, redness, irritation and dandruff after 4 weeks of application.
100% of the test persons showed significant improvements of all parameters.

ANTI-AGING FOR HAIR FOLLICLES

Hair follicles, similar to other human tissues, exhibit age related changes on a molecular level such as an altered gene expression of structural proteins. Various studies have shown a connection between cellular aging parameters in the hair follicle and oxidative stress.
H_2O_2-induced oxidative damage in the human hair follicle and shaft has been identified as a key element in hair graying which does not exclusively affect follicle melanocytes.[18]
The effects of **Ectoin®** natural on the follicular aging process were investigated in a study with follicular outer root sheath (ORS) keratinocytes and on reconstructed hair follicle models. The study showed that **Ectoin®** natural counteracts typical aging characteristics in follicular cells. The cells were subjected to oxidative stress (H_2O_2).[19]

Test protocol:

- follicular outer root sheath (ORS) keratinocytes and reconstructed hair follicle models were stressed with sub-lethal doses of hydrogen peroxide with and without pre-treatment with **Ectoin®** natural
- measured and determined parameters: expression of structural proteins and mitochondria relevant genes, amount of produced reactive oxygen species (ROS) and the degree of DNA damage (determined by COMET assay)

DNA damage in ORS keratinocytes after treatment with **Ectoin®** natural followed by hydrogen peroxide stress

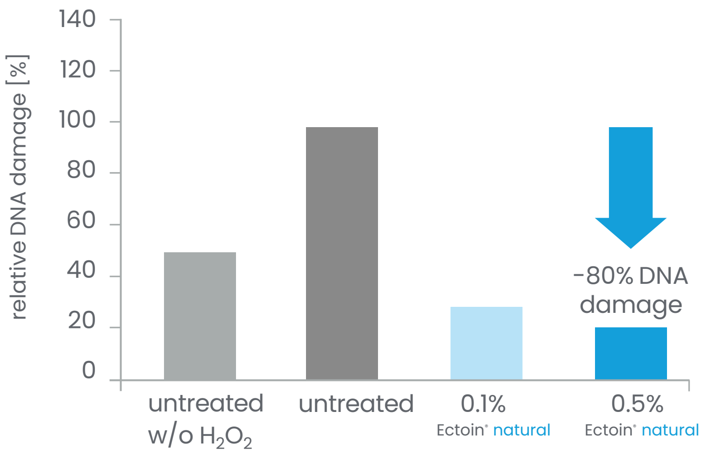


Figure 34: DNA protection in ORS keratinocytes after 24 h treatment with Ectoin® natural followed by challenging the cultures with H_2O_2 (100 µM, 20 min). Values calculated related to %tail DNA. Source: Giesen *et al.,* 2013.

Gene expression of hair keratins in reconstructed hair follicle models after treatment with **Ectoin®** natural followed by hydrogen peroxide stress

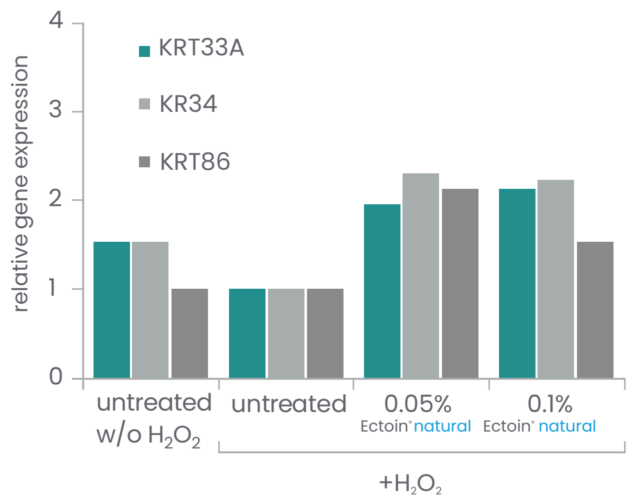


Figure 35: Keratin gene expression after 6 h treatment with Ectoin® natural followed by challenging the cultures with H_2O_2 (100 µM, 20 min). Values calculated related to the H_2O_2 control. Source: Giesen *et al.,* 2013.

Results
Ectoin® natural exhibits a significant restoration of essential cellular parameters under oxidative stress and also a significant reduction of DNA damage up to 80%.[19]
Ectoin® natural protects the cellular functions, mitochondrial DNA, and keratin synthesis from oxidative stress in hair follicles.
Furthermore, it reduces ROS in ORS keratinocytes stressed with H_2O_2. **Ectoin®** natural is able to antagonize follicular aging processes.[19]

REFERENCES

1. Heinrich U., Garbe B., Tronnier H. (2007) In vivo assessment of Ectoin: A randomized, vehicle-controlled clinical trial. SkinPharmacolPhysiol 20(4):211-8.
2. Bünger J., Degwert J., Driller H. (2001) The protective function of compatible solute Ectoin on the skin, skin cells and its biomolecules with respect to UV-radiation, immunosupression and membrane damage. IFSCC-Magazine 4(2):1-6.
3. Marini A., Reinelt K., Krutmann J., Bilstein A. (2014) Ectoine-containing cream in the treatment of mild to moderate atopic dermatitis: a randomised, comparator-controlled, intra-individual double-blind, multi-centertrial. SkinPharmacolPhysiol 27(2):57-65.
4. Kang, S., Duell, E. A., Fisher, G. J., Datta, S. C., Wang, Z. Q., Reddy, A. P., ... & Voorhees, J. J.(1995). Application of retinol to human skin in vivo induces epidermal hyperplasia and cellular retinoid binding proteins characteristic of retinoic acid but without measurable retinoic acid levels or irritation. Journal of Investigative Dermatology, 105(4), 549-556.
5. Kim, B. H., Lee, Y. S., & Kang, K. S. (2003). The mechanism of retinol-induced irritation and its application to anti-irritant development. Toxicology letters, 146(1), 65-73.
6. Graf R., Anzali S., Buenger J., Pfluecker F., Driller H. (2008) The multifunctional role of ectoine as a natural cell protectant. ClinDermatol26(4):326–333.
7. Krutmann J., Liu W., Li L., Pan X., Crawford M., Sore G., Seite S. (2014) Pollution and skin: From epidemiological and mechanistic studies to clinical implications. Journal of dermatological science76(3):163-8.
8. Unfried K., Krämer U., Sydlik U., Autengruber A., Bilstein A., Stolz S., Marini A., Schikowski T., Keymel S., Krutmann J. (2016) Reduction of neutrophilic lung inflammation by inhalation of the compatible solute ectoine: a randomized trial with elderly individuals. International JournalofChronicObstructivePulmonaryDisease11(1):2573—2583.
9. Botta C., Di Giorgio C., Sabatier A.-S., De Méo M. (2008) Genotoxicity of visible light (400–800 nm) and photoprotection assessment of ectoin, l-ergothioneine and mannitol and four sunscreens. Journal of Photochemistry and Photobiology 91(1):24-34.
10. Cho S., Shin M.H., Kim Y.K., Seo J.-E., Lee Y., Park C.-H., Chung J.H. (2009) Effects of Infrared Radiation and Heat on Human Skin Aging in vivo . Journal of Investigative Dermatology Symposium Proceedings 14: 15–19.
11. Youngson RM. Collins dictionary of human biology. Collins Publishers; 2006.
12. Godley BF, Shamsi FA, Liang FQ, Jarrett SG, Davies S, Boulton M. Blue light induces mitochondrial DNA damage and free radical production in epithelial cells. Journal of biological chemistry. 2005 Jun 3;280(22):21061-6.
13. Baldwin HE, Bhatia NC, Friedman A, Prunty T, Martin R, Seite S. The role of cutaneous microbiota harmony in maintaining a functional skin barrier. SKIN The Journal of cutaneous medicine. 2017 Oct 27;1(3.1):s139.
14. Shin et al. (2007) Reactive oxygen species produced by NADPH oxidase, xanthine oxidase, and mitochondrial electron transport system mediate heat shock-induced MMP-1 und MMP-9 expression.Free Radical Biology&Medicine,44:635-645
15. Katschinski et al. (2000) Pivotal role of reactive oxygen species as intracellular mediators of hypothermia-induced apoptosis. JBiolChem,275:21094-21098.
16. Bünger J., Driller H. (2004) Ectoin: an effective natural substance to prevent UVA-induced premature photoaging. SkinPharmacolPhysiol17 (5):232-237.
17. Grether-Beck S., Bonizzi G., Schmitt-Brenden H., Felsner I., Timmer A., Sies H., Johnson J. , Piette J., Krutmann J. (2000) Non-enzymatic triggering of the ceramide signaling cascade by solar UVA radiation. EMBOJournal19(21):5793-5800.
18. Grether-Beck S., Olaizola-Horn S., Schmitt H., Grewe M., Jahnke A., Johnson J.P., et al. (1996) Activation of transcription factor AP-2 mediates UVA radiation- and singlet oxygen-induced expression of the human intercellular adhesion molecule 1 gene. Proc Natl Acad Sci USA 93(25):14586-91.
19. Wood J. M., Decker H., Hartmann H., Chavan B., Rokos H., Spencer J. D., Hasse S., Thornton M. J., Shalbaf M., Paus R., Schallreuter K. U. (2009) Senile hair graying: H2O2-mediated oxidative stress affects human hair color by blunting methionine sulfoxide repair. The FASEB Journal 23(7):2065-75.
20. Giesen M., Fuhrmann FactsheG., et EctBriese oin® naM., tural. Rasche bitop AS., G. 03/Görlach 2022. —T., page 43 Petersohn D., Förster T. (2013) Effects of Ectoine on follicular aging processes. Henkel AG&CoKGaA, Duesseldorf, Germany.
21. Grether-Beck S, Timmer A, Felsner I, Brenden H, Brammertz D, Krutmann J. Ultraviolet A-induced signaling involves a ceramide-mediated autocrine loop leading to ceramide de novo synthesis. Journal of Investigative Dermatology. 2005 Sep 1;125(3):545-53.

Product description

100% safe
GMO-free
100% natural
halal
vegan
Ecocert
COSMOS
NaTrue
preservative free
patented
China compliant

Applications

» skin care (face, body) » sun protection » oral care
» baby care » hair care » color cosmetics

for ultra sensitive and allergic skin

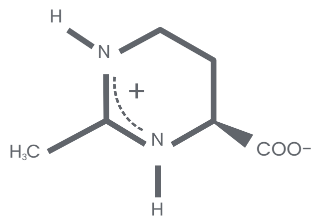
Formulation guidelines

» level of use 0.3-2%
» no incompatibility with other cosmetic active substances
» can be added directly to aqueous phase (hot or cold) prior to emulsification
» stability: pH 1-9; all temperatures (no restrictions)
» excellent water solubility (dissolves clear)

Technical data

INCI	Ectoin
Description	small cyclic amino acid derivate, 100% natural, pure molecule
Appearance	white crystalline powder, odorless
Purity	100%
Solubility	excellent solubility in water (550 g/l), Glycerin
Chemical name	(4S)-2-methyl-1,4,5,6-tetra-hydropyrimidine-4-carboxylic acid
Molecular formula	$C_6H_{10}N_2O_2$
CAS	96 702-03-3
REACH	01-0000017811-72-0001
Molar mass	142.16 g/mol
Production process	patented, bitop exclusive biotechnological fermentation
Shelf life	>5 years
Stability	pH 1-9, all temperatures, light and oxygen stable
Storage conditions	dry place, room temperature, tightly closed
Pack sizes	1 kg, 5 kg, 25 kg

Ectoin® natural
ULTIMATE PROTECTION & REPAIR



bitop
Extremolytes for life

bitop AG
Carlo-Schmid-Allee 5
44263 Dortmund
Germany

Email: sales@bitop.de
Phone: +49 231 98 77 44 0
Fax: +49 231 98 77 44 10
www.bitop.de